Supplemental Appendix for "Granular Origins of Agglomeration" by Shinnosuke Kikuchi and Daniel G. O'Connor

A Details of the Quantitative, Granular Model of Economic Geography

This section lays out the details of the quantitative model introduced in Section 3.

A.1 Environment

The country is made up of I regions, indexed by $i \in \mathcal{I} \equiv \{1, \ldots, I\}$. There is a mass ℓ of workers and a continuum of sectors $s \in [0, 1]$. The sectors produce perfectly substitutable goods but hire in distinct sectoral labor markets.

Timing. There are four periods $t \in \{-1, 0, 1, 2\}$. In period -1, workers decide where to live and then stay there for the remaining periods, determining population ℓ_i. A mass m_i of firms pay a fixed cost of the traded final good in order to enter in period 0. Each firm is then randomly assigned a sector s and gets an ex-ante productivity draw z from some known distribution.

After observing those initial productivity draws, a representative worker freely allocates her labor L_{isn} across the sectors and firms in period 1. Then, in period 2, the state of the world $\omega \in \Omega$ is revealed. This determines the short-run productivity shocks to each firm. The worker can then reallocate her labor across the firms and sectors, subject to moving costs. Firms then produce and sell their goods.

Workers. The fundamental utility of living in location i, u_i, is

$$u_i = \overline{u}_i c_i,$$

where $\overline{u}_i$ are the local amenities and c_i is consumption of the freely traded final good. Each worker has an idiosyncratic preference for each location ε_i, so that the utility the worker gets from living in location i is $u_i \varepsilon_i$. We assume that ε_i are distributed Fréchet with shape parameter $\theta > 0$.

Once the worker has decided on a location i, she needs to make her labor supply decisions. She is a risk-neutral representative agent, endowed with one unit of labor that she supplies to the market inelastically. In period 1, the worker freely allocates her units of labor across sectors and firms. In particular, she chooses her vector of labor supply $\boldsymbol{L_i} \equiv \{L_{isn}\}_{s,n}$ in the set of feasible labor allocations $\mathcal{L}$,

$$\boldsymbol{L_i} \in \mathcal{L} \equiv \left\{ \boldsymbol{L_i'} | \int_0^1 \sum_{n \in \mathcal{N}_{is}} L'_{isn} ds \leq 1 \right\}.$$

In period 2, the state of the world ω is revealed. This determines the ex-post productivity shocks for all firms. The worker then reallocates her labor across firms, choosing a vector of labor

supply $\boldsymbol{L_i(\omega)} \equiv \{L_{isn}(\omega)\}_{s,n}$ in the set of feasible labor allocations $\mathcal{L}_\Omega(\boldsymbol{L_i})$ which depends on the worker's labor choices in period 1. The set is given by

$$\mathcal{L}_\Omega(\boldsymbol{L_i}) \equiv \left\{ \boldsymbol{L_i}(\omega) | 1 = \left(\int_0^1 L_{is}^{-\frac{1}{\nu}} L_{is}(\omega)^{\frac{1+\nu}{\nu}} ds \right)^{\frac{\nu}{1+\nu}}, \right.$$
$$\left. L_{is}(\omega) = \left(\sum_{n \in \mathcal{N}_{is}} \left(\frac{L_{isn}}{L_{is}} \right)^{-\frac{1}{\kappa}} L_{isn}(\omega)^{\frac{1+\kappa}{\kappa}} \right)^{\frac{\kappa}{1+\kappa}} \right\},$$

where $L_{is} = \sum_{n \in \mathcal{N}_{is}} L_{isn}$.

Firms. Firms are the same as in the main text, so we do not restate the assumptions here.

Market Clearing. Total expected production in location i is still

$$Y_i = \mathbb{E}\left[\int_0^1 \sum_{n \in \mathcal{N}_{is}} z_{isn} a_{isn}(\omega) \ell_{isn}(\omega)^{1-\eta} ds \right].$$

The final goods are freely traded, so the market-clearing condition holds at the national level,

$$\sum_{i \in \mathcal{I}} c_i \ell_i + \psi_i m_i = \sum_{i \in \mathcal{I}} Y_i. \tag{24}$$

In the labor market, labor demand needs to equal the individual labor supplied by each worker multiplied by the number of workers. However, this needs to hold for each individual firm as labor is imperfectly substitutable across firms,

$$\ell_{isn}(\omega) = L_{isn}(\omega) \ell_i, \quad \forall s, n, \omega. \tag{25}$$

A.2 Market Structure and Equilibrium

Labor Supply Decision. We will characterize the worker's decision using backward induction, starting with the labor supply decision in periods 1 and 2, and then characterizing the migration decision in period -1 in the next section.

Conditional on living in location i, workers choose their labor allocation across firms and sectors in periods 1 and 2 to maximize their expected utility, taking wages as given. We normalize the price of the final goods to 1, so workers solve the problem.

$$\boldsymbol{L_i, L_i(\omega)} \in \underset{\boldsymbol{L_i'} \in \mathcal{L}, \boldsymbol{L_i'(\omega)} \in \mathcal{L}_\Omega(\boldsymbol{L_i'})}{\operatorname{argmax}} \mathbb{E}\left[\int_0^1 \sum_{n \in \mathcal{N}_{is}} w_{isn}(\omega) L'_{isn}(\omega) ds \right], \tag{26}$$

where $w_{isn}(\omega)$ is the equilibrium wage for firm n in sector s in state of the world ω. We will denote the maximum of (26) by w_i.

Writing out the maximization problem, we get

$$\max_{\boldsymbol{L}_i', \boldsymbol{L}_i'(\boldsymbol{\omega})} \mathbb{E}\left[\int_0^1 \sum_{n \in \mathcal{N}_{is}} w_{isn}(\omega) L_{isn}'(\omega) ds\right]$$

such that

$$\begin{aligned}
L_{is} &= \sum_{n \in \mathcal{N}_{is}} L_{isn} && (\lambda_{is}) \\
1 &= \int_0^1 L_{is} ds && (\lambda_i) \\
L_{is}(\omega)^{\frac{1+\kappa}{\kappa}} &= \sum_{n \in \mathcal{N}_{is}} \left(\frac{L_{isn}}{L_{is}}\right)^{-\frac{1}{\kappa}} L_{isn}(\omega)^{\frac{1+\kappa}{\kappa}} && (\lambda_{is}(\omega)) \\
1 &= \int_0^1 (L_{is})^{-\frac{1}{\nu}} L_{is}(\omega)^{\frac{1+\nu}{\nu}} ds. && (\lambda_i(\omega))
\end{aligned}$$

Taking the first order conditions, we find that

$$\begin{aligned}
w_{isn}(\omega) &= \lambda_{is}(\omega) \left(\frac{L_{isn}}{L_{is}}\right)^{-\frac{1}{\kappa}} \frac{1+\kappa}{\kappa} L_{isn}(\omega)^{\frac{1}{\kappa}} \\
\lambda_{is}(\omega) &= \lambda_i(\omega) (L_{is})^{-\frac{1}{\nu}} \frac{\kappa}{1+\kappa} \frac{1+\nu}{\nu} L_{is}(\omega)^{\frac{1}{\nu}-\frac{1}{\kappa}} \\
\lambda_{is} &= \mathbb{E}\left[\lambda_{is}(\omega) \frac{1}{\kappa} \left(\frac{L_{isn}}{L_{is}}\right)^{-\frac{1}{\kappa}} L_{isn}^{-1} L_{isn}(\omega)^{\frac{1+\kappa}{\kappa}}\right] \\
\lambda_i &= \lambda_{is} + \mathbb{E}\left[\lambda_i(\omega) \frac{1}{\nu} (L_{is})^{-\frac{1}{\nu}} L_{is}^{-1} L_{is}(\omega)^{\frac{1+\nu}{\nu}}\right] - \frac{1}{\kappa} L_{is}^{-1} \mathbb{E}\left[\lambda_{is}(\omega) L_{is}(\omega)^{\frac{1+\kappa}{\kappa}}\right].
\end{aligned} \tag{27}$$

One can rewrite the short-run labor supply decision in the more familiar form

$$\frac{L_{isn}(\omega)}{L_{isn}} = \left(\frac{w_{isn}(\omega)}{w_{is}(\omega)}\right)^{\kappa} \frac{L_{is}(\omega)}{L_{is}}, \quad \frac{L_{is}(\omega)}{L_{is}} = \left(\frac{w_{is}(\omega)}{w_i(\omega)}\right)^{\nu},$$

where

$$w_{is}(\omega) \equiv \left(\sum_{n \in \mathcal{N}_{is}} \frac{L_{isn}}{L_{is}} w_{isn}(\omega)^{1+\kappa}\right)^{\frac{1}{1+\kappa}}, \quad w_i(\omega) \equiv \left(\int_0^1 L_{is} w_{is}(\omega)^{1+\nu} ds\right)^{\frac{1}{1+\nu}}.$$

Transforming the per capita labor supply, $L_{isn}(\omega)$, to the total labor supplied $\ell_{isn}(\omega)$ and taking logs implies,

$$\log \ell_{isn}(\omega) - \log \ell_{isn} = \kappa \log w_{isn}(\omega) - (\kappa - \nu) \log w_{is}(\omega) - \nu \log w_i(\omega). \tag{28}$$

Including firm-year specific amenities implies the equation (14).

Migration Decision. Each worker chooses the location that maximizes their utility. Therefore, the population in location i satisfies

$$\ell_i = \int_{\mathbb{R}^I} \mathbb{1}_{i \in \text{argmax}_{i'} \overline{u}_{i'} w_{i'}} dG(\varepsilon) \cdot \ell, \tag{29}$$

where G is the joint distribution of ε. This is a standard problem in the literature with a Fréchet distribution. It implies that $\ell_i = (u_i/u)^\theta \cdot \ell$, where $u = (\sum_i (u_i)^\theta)^{\frac{1}{\theta}}$.

Labor Demand - Competitive. We will consider three different conduct assumptions on firms after they enter. The first assumption is that they are competitive. Then each active firm maximizes profits, taking wages and prices as given,

$$\ell_{isn}(\omega) \in \underset{\ell'}{\text{argmax}} \quad z_{isn} a_{isn}(\omega) (\ell')^{1-\eta} - w_{isn}(\omega) \ell'. \tag{30}$$

This implies that

$$z_{isn} a_{isn}(\omega) \ell_{isn}(\omega)^{-\eta} = w_{isn}(\omega). \tag{31}$$

Labor Demand - Cournot. Under Cournot competition, the firm takes as given the labor decisions of the other firms in its own sector. We assume that the firm then takes as given the workers' other options in other sectors. In the math, that will imply that the firm will take $\lambda_i(\omega)$ and λ_i in equation (27) as given. Combining some of the first-order necessary conditions of the worker's problem, we can write the firm problem, using $\boldsymbol{x}_{isn} \equiv \{w_{isn}(\omega), \ell_{isn}(\omega), \ell_{is}(\omega), \ell_{isn}, \ell_{is}\}$, as,

$$\begin{aligned}
\boldsymbol{x}_{isn} \in \underset{\boldsymbol{x}'_{isn}}{\text{argmax}} \quad & \mathbb{E}\left[z_{isn} a_{isn}(\omega) \ell'_{isn}(\omega)^{1-\eta} - w'_{isn}(\omega) \ell'_{isn}(\omega) \right] \\
\text{s.t.} \quad & (\ell'_{is})^{-\frac{1}{\kappa}} \ell'_{is}(\omega)^{\frac{1+\kappa}{\kappa}} = \left(\ell'_{isn}\right)^{-\frac{1}{\kappa}} \ell'_{isn}(\omega)^{\frac{1+\kappa}{\kappa}} + \sum_{n' \neq n} (\ell_{isn'})^{-\frac{1}{\kappa}} \ell_{isn'}(\omega)^{\frac{1+\kappa}{\kappa}} \\
& \ell'_{is} = \ell'_{isn} + \sum_{n' \neq n} \ell_{isn'} \\
& w'_{isn}(\omega) = \lambda_i(\omega) \frac{1+\nu}{\nu} \left(\frac{\ell_{is}(\omega)}{\ell_{is}} \right)^{\frac{1}{\nu} - \frac{1}{\kappa}} \left(\frac{\ell_{isn}(\omega)}{\ell_{isn}} \right)^{\frac{1}{\kappa}} \\
& \lambda_i (1+\kappa)\nu = \mathbb{E}\left[\lambda_i(\omega)(1+\nu) \left(\frac{\ell_{is}(\omega)}{\ell_{is}} \right)^{\frac{1}{\nu} - \frac{1}{\kappa}} \left(\frac{\ell_{isn}(\omega)}{\ell_{isn}} \right)^{\frac{1+\kappa}{\kappa}} \right] \\
& \qquad + \mathbb{E}\left[\lambda_i(\omega)(\kappa - \nu) \left(\frac{\ell_{is}(\omega)}{\ell_{is}} \right)^{\frac{1+\nu}{\nu}} \right],
\end{aligned} \tag{32}$$

where we have transformed the per capita variables into the total amount of labor.

Taking the first order conditions and simplifying gives the following necessary conditions,

$$
\begin{aligned}
0 = {} & \mathbb{E}[w_{isn}(\omega)\ell_{isn}(\omega)] - \ell_{isn}\mathbb{E}\left[\Lambda_{isn}(\omega)\left(\left(\frac{\ell_{isn}(\omega)}{\ell_{isn}}\right)^{\frac{1+\kappa}{\kappa}} - \left(\frac{\ell_{is}(\omega)}{\ell_{is}}\right)^{\frac{1+\kappa}{\kappa}}\right)\right] \\
& - (1-\eta)\mathbb{E}\left[z_{isn}a_{isn}(\omega)\ell_{isn}(\omega)^{1-\eta}\right] \\
0 = {} & \frac{1+\kappa}{\kappa}w_{isn}(\omega)\ell_{isn}(\omega) - \frac{1+\kappa}{\kappa}\Lambda_{isn}(\omega)\ell_{isn}(\omega)\left(\frac{\ell_{isn}(\omega)}{\ell_{isn}}\right)^{\frac{1}{\kappa}} \\
& - \frac{\Lambda^w_{isn}}{\lambda_i(\omega)}\frac{1+\kappa}{\kappa}\nu w_{isn}(\omega)\frac{\ell_{isn}(\omega)}{\ell_{isn}} - (1-\eta)z_{isn}a_{isn}(\omega)\ell_{isn}(\omega)^{1-\eta} \\
0 = {} & -\left(\frac{1}{\nu}-\frac{1}{\kappa}\right)\left[w_{isn}(\omega)\ell_{isn}(\omega) - \frac{\Lambda^w_{isn}}{\lambda_i(\omega)}\nu w_{isn}(\omega)\frac{\ell_{isn}(\omega)}{\ell_{isn}}\right. \\
& \left. - \kappa(1+\nu)\Lambda^w_{isn}\left(\frac{\ell_{is}(\omega)}{\ell_{is}}\right)^{\frac{1+\nu}{\nu}}\right] - \Lambda_{isn}(\omega)\ell_{is}\frac{1+\kappa}{\kappa}\left(\frac{\ell_{is}(\omega)}{\ell_{is}}\right)^{\frac{1+\kappa}{\kappa}}
\end{aligned} \tag{33}
$$

Labor Demand - Bertrand. Under Bertrand competition, the firm takes as given the wage decisions of the other firms in its own sector. We assume that the firm then takes as given the workers' other options in other sectors. Just as for Cournot, that implies that the firm takes $\lambda_i(\omega)$ and λ_i in equation (27) as given. Combining some of the first order necessary conditions of the worker's problem, we can write the firm problem, using $x_{isn} \equiv \{w_{isn}(\omega), w_{is}(\omega), \ell_{isn}(\omega), \ell_{isn}, \ell_{is}\}$, as,

$$
\begin{aligned}
\boldsymbol{x}_{isn} \in \underset{\boldsymbol{x}'_{isn}}{\text{argmax}} \quad & \mathbb{E}\left[z_{isn}a_{isn}(\omega)\ell'_{isn}(\omega)^{1-\eta} - w'_{isn}(\omega)\ell'_{isn}(\omega)\right] \\
\text{s.t.} \quad & w'_{is}(\omega)^{1+\kappa} = \frac{\ell'_{isn}}{\ell'_{is}}w'_{isn}(\omega)^{1+\kappa} + \sum_{n'\neq n}\frac{\ell'_{isn'}}{\ell'_{is}}w'_{isn'}(\omega)^{1+\kappa} \\
& \ell'_{is} = \ell'_{isn} + \sum_{n'\neq n}\ell_{isn'} \\
& \ell_{isn}(\omega)' = \lambda_i(\omega)^{-\nu}\left(\frac{\nu}{1+\nu}\right)\frac{\ell'_{isn}}{\ell'_i}w'_{is}(\omega)^{\nu-\kappa}w'_{isn}(\omega)^{\kappa} \\
& \lambda_i = \lambda_i(\omega)^{-\nu}\frac{1}{1+\kappa}\left(\frac{\nu}{1+\nu}\right)^{\nu}\ell_i^{-1}\mathbb{E}\left[w'_{is}(\omega)^{\nu\kappa\left(\frac{1}{\kappa}-\frac{1}{\nu}\right)}w'_{isn}(\omega)^{1+\kappa}\right] \\
& \qquad + \lambda_i(\omega)^{-\nu}\frac{\kappa}{1+\kappa}\left(\frac{1}{\nu}-\frac{1}{\kappa}\right)\left(\frac{\nu}{1+\nu}\right)^{1+\nu}\ell_i^{-1}\mathbb{E}\left[w'_{is}(\omega)^{1+\nu}\right],
\end{aligned} \tag{34}
$$

where we have transformed the per capita variables into the total amount of labor.

Taking the first order conditions and simplifying gives the following necessary conditions,

$$
\begin{aligned}
0 = {} & \mathbb{E}\left[w_{isn}(\omega)\ell_{isn}(\omega)\right] - \ell_{isn}\mathbb{E}\left[\Lambda_{isn}(\omega)\left(w_{isn}(\omega)^{1+\kappa} - w_{is}(\omega)^{1+\kappa}\right)\right] \\
& - (1-\eta)\mathbb{E}\left[z_{isn}a_{isn}(\omega)\ell_{isn}(\omega)^{1-\eta}\right]
\end{aligned}
$$

$$
\begin{aligned}
0 = & \frac{1+\kappa}{\kappa} w_{isn}(\omega)\ell_{isn}(\omega) - \frac{1+\kappa}{\kappa}\Lambda_{isn}(\omega)\ell_{isn} w_{isn}(\omega)^{1+\kappa} \\
& - \frac{\Lambda^w_{isn}}{\kappa}\lambda_i(\omega)^{-\nu}\left(\frac{\nu}{1+\nu}\right)^{\nu}\ell_i^{-1} w_{is}(\omega)^{\nu-\kappa} w_{isn}(\omega)^{1+\kappa} \\
& - (1-\eta) z_{isn} a_{isn}(\omega)\ell_{isn}(\omega)^{1-\eta} \\
0 = & -\frac{\kappa-\nu}{\kappa}\left[w_{isn}(\omega)\ell_{isn}(\omega) - \frac{1}{1+\kappa}\Lambda^w_{isn}\lambda_i(\omega)^{-\nu}\left(\frac{\nu}{1+\nu}\right)^{\nu}\ell_i^{-1} w_{is}(\omega)^{\nu-\kappa} w_{isn}(\omega)^{1+\kappa}\right. \\
& \left. - \Lambda^w_{isn}\lambda(\omega)^{-\nu}\frac{\kappa}{1+\kappa}\left(\frac{\nu}{1+\nu}\right)^{\nu}\ell_i^{-1} w_{is}(\omega)^{1+\nu}\right] \\
& - \frac{1+\kappa}{\kappa}\Lambda_{isn}(\omega)\left(\kappa\ell_{is} w_{is}(\omega)^{1+\kappa} + (\nu-\kappa)\ell_{isn} w_{isn}(\omega)^{1+\kappa}\right)
\end{aligned} \tag{35}
$$

Entry. Entry is the same as in the baseline model. Thus, free entry implies that average profits are equal to the cost of entry, (5).

A.3 Characterizing Cournot Competition

In this subsection, we characterize the equilibrium under Cournot Competition. The expression for production given in Lemma 8 holds no matter how firms behave. Therefore, we go through and characterize what total wage payments are, which then implies profits.

Taking a second-order approximation to total wage payments and the firm FOCs associated with Cournot competition implies the next proposition.

Lemma 4. *If firms compete à la Cournot, total wage compensation in location i can be written,*

$$
w_i\ell_i = (1-\eta) z_i(m_i)^{\eta}(\ell_i)^{1-\eta}\left(\tilde{\Phi}^c(m) + \Psi^c(m)\right),
$$

where $\tilde{\Phi}^c(m)$ is defined as in Lemma 8 and

$$
\Psi^c(m) \equiv \frac{1+\kappa}{\kappa}\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\mathbb{E}\left[\frac{\hat{\Lambda}_{isn}(\omega)}{w_i}\left(\hat{\ell}_{isn}(\omega) - \hat{\ell}_{isn} - \hat{\ell}_{is}(\omega) + \hat{\ell}_{is}\right)\right] ds.
$$

Proof. First, we note that because workers are freely mobile across all sectors in period 1, the equilibrium with no ex-post shocks and Cournot competition is the same as the equilibrium with no ex-post shocks and competitive firms. Doing a second-order approximation around the point with no ex-post shocks implies

$$
\begin{aligned}
& \mathbb{E}\left[\int_0^1 \sum_{n\in\mathcal{N}_{is}} w_{isn}(\omega)\ell_{isn}(\omega) ds\right] \\
& = \overline{w}_i\overline{\ell}_i\left(\int_0^1 \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\ell_i}\left(1 + \mathbb{E}[\hat{w}_{isn}(\omega)] + \mathbb{E}[\hat{\ell}_{isn}(\omega)] + \frac{1}{2}\mathbb{E}[(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega))^2]\right) ds\right)
\end{aligned}
$$

We then take a second-order log approximation to the firm FOCs in equation (33). However, with no shocks, $\Lambda_{isn}(\omega) = 0$, so we leave that in levels. The first equation becomes

$$\begin{aligned}\mathbb{E}[\hat{a}_{isn}(\omega)] + (1-\eta)\mathbb{E}[\hat{\ell}_{isn}(\omega)] + \frac{1}{2}\mathbb{E}\left[(\hat{a}_{isn}(\omega) + (1-\eta)\hat{\ell}_{isn}(\omega))^2\right] \\ = \mathbb{E}[\hat{w}_{isn}(\omega)] + \mathbb{E}[\hat{\ell}_{isn}(\omega)] + \frac{1}{2}\mathbb{E}\left[\left(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega)\right)^2\right] \\ - \frac{1+\kappa}{\kappa}\mathbb{E}\left[\frac{\hat{\Lambda}_{isn}(\omega)}{w_i}\left(\hat{\ell}_{isn}(\omega) - \hat{\ell}_{isn} - \hat{\ell}_{is}(\omega) + \hat{\ell}_{is}\right)\right].\end{aligned}$$

We also need the second-order approximation to the labor constraints embedded in $\mathcal{L}$ and $\mathcal{L}_\Omega(\cdot)$,

$$\begin{aligned}\hat{\ell}_{is}(\omega) + \frac{1}{2}\frac{\kappa}{1+\kappa}\left(-\frac{1}{\kappa}\hat{\ell}_{is} + \frac{1+\kappa}{\kappa}\hat{\ell}_{is}(\omega)\right)^2 + \frac{1}{1+\kappa}\hat{\ell}_{is}^2 \\ = \sum_{n\in\mathcal{N}_{is}}\frac{\ell_{isn}}{\ell_{is}}\left(\hat{\ell}_{isn}(\omega) + \frac{1}{2}\frac{\kappa}{1+\kappa}\left(-\frac{1}{\kappa}\hat{\ell}_{isn} + \frac{1+\kappa}{\kappa}\hat{\ell}_{isn}(\omega)\right)^2 + \frac{1}{1+\kappa}\hat{\ell}_{isn}^2\right) \\ 0 = \int_0^1 \ell_{is}\left(\hat{\ell}_{is}(\omega) + \frac{1}{2}\frac{\nu}{1+\nu}\left(-\frac{1}{\nu}\hat{\ell}_{is} + \frac{1+\nu}{\nu}\hat{\ell}_{is}(\omega)\right)^2 + \frac{1}{1+\nu}\hat{\ell}_{is}^2\right)ds.\end{aligned}$$

So then we can write

$$\begin{aligned}\int_0^1 \frac{\ell_{is}}{\ell_i}\sum_{n\in\mathcal{N}_{is}}\frac{\ell_{isn}}{\ell_{is}}\left(\mathbb{E}\left[\hat{w}_{isn}(\omega)\right] + \mathbb{E}\left[\hat{\ell}_{isn}(\omega)\right] + \frac{1}{2}\mathbb{E}[(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega))^2]\right)ds \\ = \frac{1}{2}\int_0^1 \frac{\ell_{is}}{\ell_i}\sum_{n\in\mathcal{N}_{is}}\frac{\ell_{isn}}{\ell_{is}}\mathbb{E}\left[(\hat{a}_{isn}(\omega) + (1-\eta)\hat{\ell}_{isn}(\omega))^2\right]ds \\ + \frac{1+\kappa}{\kappa}\int_0^1 \frac{\ell_{is}}{\ell_i}\sum_{n\in\mathcal{N}_{is}}\frac{\ell_{isn}}{\ell_{is}}\mathbb{E}\left[\frac{\hat{\Lambda}_{isn}(\omega)}{w_i}\left(\hat{\ell}_{isn}(\omega) - \hat{\ell}_{isn} - \hat{\ell}_{is}(\omega) + \hat{\ell}_{is}\right)\right]ds \\ + \int_0^1 \frac{\ell_{is}}{\ell_i}\sum_{n\in\mathcal{N}_{is}}\frac{\ell_{isn}}{\ell_{is}}\left(\mathbb{E}[\hat{a}_{isn}(\omega)] + (1-\eta)\mathbb{E}[\hat{\ell}_{isn}(\omega)]\right)ds \\ = \mathbb{E}[\hat{a}_{isn}(\omega)] + \frac{1}{2}\mathbb{E}[\hat{a}_{isn}(\omega)^2] + (1-\eta)\int_0^1 \frac{\ell_{is}}{\ell_i}\sum_{n\in\mathcal{N}_{is}}\frac{\ell_{isn}}{\ell_{is}}\mathbb{E}[\hat{a}_{isn}(\omega)\hat{\ell}_{isn}(\omega)]ds \\ - \eta\frac{1-\eta}{2}\int_0^1 \frac{\ell_{is}}{\ell}\sum_{n\in\mathcal{N}_{is}}\frac{\ell_{isn}}{\ell_{is}}\mathbb{E}[\hat{\ell}_{isn}(\omega)^2]ds \\ - \frac{1}{\kappa}\frac{1-\eta}{2}\int_0^1 \frac{\ell_s}{\ell}\sum_{n\in\mathcal{N}_s}\frac{\ell_{isn}}{\ell_{is}}\mathbb{E}\left[\left(\hat{\ell}_{sn}(\omega) - \hat{\ell}_{isn}\right)^2\right]ds \\ - \left(\frac{1}{\nu} - \frac{1}{\kappa}\right)\frac{1-\eta}{2}\int_0^1 \frac{\ell_{is}}{\ell}\mathbb{E}\left[\left(\hat{\ell}_{is}(\omega) - \hat{\ell}_{is}\right)^2\right]ds \\ + \frac{1+\kappa}{\kappa}\int_0^1 \frac{\ell_{is}}{\ell_i}\sum_{n\in\mathcal{N}_{is}}\frac{\ell_{isn}}{\ell_{is}}\mathbb{E}\left[\frac{\hat{\Lambda}_{isn}(\omega)}{w_i}\left(\hat{\ell}_{isn}(\omega) - \hat{\ell}_{isn} - \hat{\ell}_{is}(\omega) + \hat{\ell}_{is}\right)\right]ds.\end{aligned}$$

This completes the proof. □

Lemma 4 gives wages completely in second-order terms. Therefore, following Benigno and Woodford (2003) and Benigno and Woodford (2012), we can then approximate the labor constraints and firm first-order conditions with a log first-order approximation. We then computationally evaluate $\tilde{\Phi}^c(m)$ and $\Psi^c(m)$ and use those functions in calculating the equilibrium.

A.4 Characterizing Bertrand Competition

In this subsection, we characterize the equilibrium under Bertrand Competition. The expression for production given in Lemma 8 holds no matter how firms behave. Therefore, we go through and characterize what total wage payments are, which then implies profits. We also show that in the limit as $\kappa \to \infty$, Bertrand competition converges to the perfect competition case. Thus, the original model in Section 2 is equivalent to a model with Bertrand competition.

Taking a second-order approximation to total wage payments and the firm FOCs associated with Bertrand competition implies the next proposition.

Lemma 5. *If firms compete à la Bertrand, total wage compensation in location i can be written,*

$$w_i \ell_i = (1-\eta) z_i (m_i)^{\eta} (\ell_i)^{1-\eta} \left(\tilde{\Phi}^b(m) + \Psi^b(m) \right),$$

where $\tilde{\Phi}^b(m)$ is defined as in Lemma 8 and

$$\Psi^b(m) \equiv -(1+\kappa) w_i^{\kappa} \int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathbb{E}\left[\hat{\Lambda}_{isn}(\omega) \left(\hat{w}_{isn}(\omega) - \hat{w}_{is}(\omega) \right) \right] ds.$$

Proof. First, we note that because workers are freely mobile across all sectors in period 1, the equilibrium with no ex-post shocks and Bertrand competition is the same as the equilibrium with no ex-post shocks and competitive firms. Doing a second-order approximation around the point with no ex-post shocks implies,

$$\begin{aligned} &\mathbb{E}\left[\int_0^1 \sum_{n \in \mathcal{N}_{is}} w_{isn}(\omega) \ell_{isn}(\omega) ds \right] \\ &= \overline{w}_i \overline{\ell}_i \left(\int_0^1 \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\ell_i} \left(1 + \mathbb{E}[\hat{w}_{isn}(\omega)] + \mathbb{E}[\hat{\ell}_{isn}(\omega)] + \frac{1}{2} \mathbb{E}[(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega))^2] \right) ds \right) \end{aligned}$$

We then take a second-order log approximation to the firm FOCs in equation (35). However, with no shocks, $\Lambda_{isn}(\omega) = 0$, so we leave that in levels. The first equation becomes

$$\mathbb{E}[\hat{a}_{isn}(\omega)] + (1-\eta)\mathbb{E}[\hat{\ell}_{isn}(\omega)] + \frac{1}{2}\mathbb{E}\left[\left(\hat{a}_{isn}(\omega) + (1-\eta)\hat{\ell}_{isn}(\omega) \right)^2 \right]$$

$$= \mathbb{E}[\hat{w}_{isn}(\omega)] + \mathbb{E}[\hat{\ell}_{isn}(\omega)] + \frac{1}{2}\mathbb{E}\left[\left(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega)\right)^2\right]$$
$$+ (1+\kappa) w_i^{\kappa} \mathbb{E}\left[\hat{\Lambda}_{isn}(\omega)\left(\hat{w}_{isn}(\omega) - \hat{w}_{is}(\omega)\right)\right].$$

We also need the second-order approximation to the labor constraints embedded in $\mathcal{L}$ and $\mathcal{L}_{\Omega}(\cdot)$,

$$\hat{\ell}_{is}(\omega) + \frac{1}{2}\frac{\kappa}{1+\kappa}\left(-\frac{1}{\kappa}\hat{\ell}_{is} + \frac{1+\kappa}{\kappa}\hat{\ell}_{is}(\omega)\right)^2 + \frac{1}{1+\kappa}\hat{\ell}_{is}^2$$
$$= \sum_{n\in\mathcal{N}_{is}} \frac{\ell_{isn}}{\overline{\ell}_{is}}\left(\hat{\ell}_{isn}(\omega) + \frac{1}{2}\frac{\kappa}{1+\kappa}\left(-\frac{1}{\kappa}\hat{\ell}_{isn} + \frac{1+\kappa}{\kappa}\hat{\ell}_{isn}(\omega)\right)^2 + \frac{1}{1+\kappa}\hat{\ell}_{isn}^2\right)$$
$$0 = \int_0^1 \overline{\ell}_{is}\left(\hat{\ell}_{is}(\omega) + \frac{1}{2}\frac{\nu}{1+\nu}\left(-\frac{1}{\nu}\hat{\ell}_{is} + \frac{1+\nu}{\nu}\hat{\ell}_{is}(\omega)\right)^2 + \frac{1}{1+\nu}\hat{\ell}_{is}^2\right) ds.$$

So then we can write

$$\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\left(\mathbb{E}\left[\hat{w}_{isn}(\omega)\right] + \mathbb{E}\left[\hat{\ell}_{isn}(\omega)\right] + \frac{1}{2}\mathbb{E}[(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega))^2]\right) ds$$
$$= \frac{1}{2}\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathbb{E}\left[(\hat{a}_{isn}(\omega) + (1-\eta)\hat{\ell}_{isn}(\omega))^2\right] ds$$
$$- (1+\kappa) w_i^{\kappa} \int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathbb{E}\left[\hat{\Lambda}_{isn}(\omega)\left(\hat{w}_{isn}(\omega) - \hat{w}_{is}(\omega)\right)\right] ds$$
$$+ \int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\left(\mathbb{E}[\hat{a}_{isn}(\omega)] + (1-\eta)\mathbb{E}[\hat{\ell}_{isn}(\omega)]\right) ds$$
$$= \mathbb{E}[\hat{a}_{isn}(\omega)] + \frac{1}{2}\mathbb{E}[\hat{a}_{isn}(\omega)^2] + (1-\eta)\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathbb{E}[\hat{a}_{isn}(\omega)\hat{\ell}_{isn}(\omega)] ds$$
$$- \eta\frac{1-\eta}{2}\int_0^1 \frac{\overline{\ell}_{is}}{\ell} \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathbb{E}[\hat{\ell}_{isn}(\omega)^2] ds$$
$$- \frac{1}{\kappa}\frac{1-\eta}{2}\int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n\in\mathcal{N}_s} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathbb{E}\left[\left(\hat{\ell}_{sn}(\omega) - \hat{\ell}_{isn}\right)^2\right] ds$$
$$- \left(\frac{1}{\nu} - \frac{1}{\kappa}\right)\frac{1-\eta}{2}\int_0^1 \frac{\overline{\ell}_{is}}{\ell} \mathbb{E}\left[\left(\hat{\ell}_{is}(\omega) - \hat{\ell}_{is}\right)^2\right] ds$$
$$- (1+\kappa)\overline{w}_i^{\kappa} \int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathbb{E}\left[\hat{\Lambda}_{isn}(\omega)\left(\hat{w}_{isn}(\omega) - \hat{w}_{is}(\omega)\right)\right] ds.$$

This completes the proof. □

Lemma 5 gives wages completely in second-order terms. Therefore, following Benigno and Woodford (2003) and Benigno and Woodford (2012), we can then approximate the labor constraints and firm first-order conditions with a log first-order approximation. We then computationally evaluate $\tilde{\Phi}^b(m)$ and $\Upsilon^b(m)$ and use those functions in calculating the equilibrium.

Next, we show the equivalence to the competitive equilibrium as $\kappa \to \infty$.

Lemma 6. *In the limit as $\kappa \to \infty$, labor and profits are the same under Bertrand competition and perfect competition.*

Proof. The system is characterized by the constraints in the firm maximization problem (34) and the FOCs (35). We start by solving the system in the absence of shocks and then doing a first order approximation around that point.

In the absence of shocks, wages are equalized across all firms. We will denote this wage by $\overline{w}_i$. Solving the system then implies that

$$\overline{\lambda}_i(\omega)^{-\nu} = \frac{1+\nu}{\nu}\ell_i\overline{w}_i^{-\nu} \qquad \overline{w}_i = (1-\eta)z_{isn}\overline{\ell}_{isn}^{-\eta}$$

$$\overline{\Lambda}_{isn}(\omega) = 0 \qquad \overline{\lambda}_i = \frac{1}{\nu}\left(\frac{\nu}{1+\nu}\right)^{\nu}\overline{w}_i$$

$$\overline{\ell}_{isn} = \overline{\Lambda}^w_{isn}\left(\frac{\nu}{1+\nu}\right)^{\nu-1}.$$

Taking the log first order approximation implies the system of equations,

$$\begin{aligned}
\hat{\ell}_{is} + (1+\kappa)\hat{w}_{is}(\omega) &= \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\left(\hat{\ell}_{isn} + (1+\kappa)\hat{w}_{isn}(\omega)\right) \\
\hat{\ell}_{isn}(\omega) &= -\nu\hat{\lambda}_i(\omega) + \hat{\ell}_{isn} + (\nu-\kappa)\hat{w}_{is}(\omega) + \kappa\hat{w}_{isn}(\omega) \\
\frac{1}{\nu}\left(\frac{\nu}{1+\nu}\right)^{\nu}\hat{\lambda}_i &= \frac{1}{1+\kappa}\mathbb{E}\left[-\nu\hat{\lambda}_i(\omega) + (\nu-\kappa)\hat{w}_{is}(\omega) + (1+\kappa)\hat{w}_{isn}(\omega)\right] \\
&\quad + \frac{\kappa}{1+\kappa}\left(\frac{1}{\nu}-\frac{1}{\kappa}\right)\left(\frac{\nu}{1+\nu}\right)^{\nu}\mathbb{E}[-\nu\hat{\lambda}_i(\omega) + (1+\nu)\hat{w}_{is}(\omega)] \\
\mathbb{E}[\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega)] &= \mathbb{E}[\hat{a}_{isn}(\omega) + (1-\eta)\hat{\ell}_{isn}(\omega)] \\
\frac{1+\kappa}{\kappa}\left(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega)\right) &= \frac{1+\kappa}{\kappa}\overline{w}_i^{\kappa}\hat{\Lambda}_{isn}(\omega) \\
&\quad + \frac{1}{\kappa}\left(\hat{\Lambda}^w_{isn} - \nu\hat{\lambda}_i(\omega) + (\nu-\kappa)\hat{w}_{is}(\omega) + (1+\kappa)\hat{w}_{isn}(\omega)\right) \\
&\quad + \hat{a}_{isn}(\omega) + (1-\eta)\hat{\ell}_{isn}(\omega) \\
0 &= -\frac{1+\kappa}{\kappa}\left(\kappa\overline{\ell}_{is} + (\nu-\kappa)\overline{\ell}_{isn}\right)\overline{w}_i^{\kappa}\hat{\Lambda}_{isn}(\omega) \\
&\quad - \frac{\kappa-\nu}{\kappa}\overline{\ell}_{isn}\left(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega)\right) \\
&\quad + \frac{\kappa-\nu}{\kappa}\frac{1}{1+\kappa}\overline{\ell}_{isn}\left(\hat{\Lambda}^w_{isn} - \nu\hat{\lambda}_i(\omega) + (\nu-\kappa)\hat{w}_{is}(\omega) + (1+\kappa)\hat{w}_{isn}(\omega)\right) \\
&\quad + \frac{\kappa-\nu}{\nu}\frac{\kappa}{1+\kappa}\overline{\ell}_{isn}\left(\hat{\Lambda}^w_{isn} - \nu\hat{\lambda}_i(\omega) + (1+\nu)\hat{w}_{is}(\omega)\right).
\end{aligned}$$

We also have a first order approximation to the labor market clearing conditions,

$$\hat{\ell}_{is}(\omega) = \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{\ell}_{isn}(\omega), \qquad 0 = \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \hat{\ell}_{is}(\omega) ds,$$

$$\hat{\ell}_{is} = \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{\ell}_{isn}, \qquad 0 = \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \hat{\ell}_{is} ds.$$

Note that $\mathbb{E}[\hat{a}_{isn}(\omega)] = 0$, so $\mathbb{E}[\hat{w}_{isn}(\omega)] = -\eta \mathbb{E}[\hat{\ell}_{isn}(\omega)]$. Taking the second equation and summing across firms within a sector gives

$$\sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{\ell}_{isn}(\omega) = -\nu \hat{\lambda}_i(\omega) + \hat{\ell}_{is} + \nu \hat{w}_{is}(\omega).$$

Aggregating across sectors, and using the fact that $0 = \int_0^1 \overline{\ell}_{is} \hat{\ell}_{is} ds$,

$$\int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{\ell}_{isn}(\omega) ds = -\nu \hat{\lambda}_i(\omega) + \nu \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \hat{w}_{is}(\omega) ds$$

Then, $\hat{\lambda}_i(\omega)$ is constant. Taking expectations, and using the fact that $\mathbb{E}[\hat{w}_{isn}(\omega)] = -\eta \mathbb{E}[\hat{\ell}_{isn}(\omega)]$,

$$0 = (1 + \nu\eta) \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathbb{E}[\hat{\ell}_{isn}(\omega)] ds = -\nu \hat{\lambda}_i(\omega).$$

Therefore, $\mathbb{E}[\hat{\ell}_{isn}(\omega)] = \mathbb{E}[\hat{w}_{isn}(\omega)] = 0$, and we also get $\hat{\ell}_{isn} = 0$. This further implies that $\hat{\lambda}_i = 0$. Rewriting the last two equations gives

$$\hat{w}_{isn}(\omega) + \eta \hat{\ell}_{isn}(\omega) = \frac{1+\kappa}{\kappa} \overline{w}_i^{\kappa} \hat{\Lambda}_{isn}(\omega) + \frac{1}{\kappa} \hat{\Lambda}^w_{isn} + \hat{a}_{isn}(\omega)$$

$$\frac{1+\kappa}{\kappa} \left(\kappa \overline{\ell}_{is} + (\nu - \kappa) \overline{\ell}_{isn} \right) \frac{\overline{w}_i^{\kappa}}{\overline{\ell}_{isn}} \hat{\Lambda}_{isn}(\omega) = \frac{\kappa - \nu}{\kappa} \frac{\kappa}{1+\kappa} \left(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega) - (1+\nu) \hat{w}_{is}(\omega) \right) + \frac{\kappa - \nu}{1+\kappa} \frac{1}{\kappa} \hat{\Lambda}^w_{isn}.$$

Taking expectations of the first equation implies that

$$\frac{1+\kappa}{\kappa} \overline{w}_i^{\kappa} \mathbb{E}[\hat{\Lambda}_{isn}(\omega)] = -\frac{1}{\kappa} \hat{\Lambda}^w_{isn}.$$

Then taking the expectations of the second equation implies that $\hat{\Lambda}^w_{isn} = 0$. Substituting $\hat{\ell}_{isn}(\omega) = (\nu - \kappa) \hat{w}_{is}(\omega) + \kappa \hat{w}_{isn}(\omega)$ into the second equation gives

$$\left(\kappa + (\nu - \kappa) \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \right) \frac{1+\kappa}{\kappa} \overline{w}_i^{\kappa} \hat{\Lambda}_{isn}(\omega) = \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} (\kappa - \nu) (\hat{w}_{isn}(\omega) - \hat{w}_{is}(\omega)).$$

Substituting this into the first equation,

$$\hat{w}_{isn}(\omega) + \eta(\nu-\kappa)\hat{w}_{is}(\omega) + \eta\kappa\hat{w}_{isn}(\omega) = \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa + (\nu-\kappa)\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}} (\hat{w}_{isn}(\omega) - \hat{w}_{is}(\omega)) + \hat{a}_{isn}(\omega).$$

Solving for $\hat{w}_{isn}(\omega)$,

$$\hat{w}_{isn}(\omega) = \frac{\hat{a}_{isn}(\omega)}{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}} + (\kappa-\nu)\frac{\eta - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{1}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}}{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}} \hat{w}_{is}(\omega).$$

Thus, in the limit as $\kappa \to \infty$, $\hat{w}_{isn}(\omega) = \hat{w}_{is}(\omega)$. Then the sectoral wages are

$$\begin{aligned}
\hat{w}_{is}(\omega) &= \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{w}_{isn}(\omega) \\
&= \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\hat{a}_{isn}(\omega)}{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}} \\
&\qquad + \hat{w}_{is}(\omega) \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\eta(\kappa-\nu) - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}}{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}} \\
\hat{w}_{is}(\omega) &= \frac{\sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\hat{a}_{isn}(\omega)}{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}}}{1 - \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\eta(\kappa-\nu) - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}}{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}}} \\
&= \frac{\sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\hat{a}_{isn}(\omega)}{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}}}{\sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}} - \eta(\kappa-\nu) + \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}}{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}}} \\
&= \frac{\sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\hat{a}_{isn}(\omega)}{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}}}{\sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{1+\eta\nu}{1+\eta\kappa - \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}}}.
\end{aligned}$$

Taking the limit as $\kappa \to \infty$, $\hat{w}_{is}(\omega) = \frac{1}{1+\eta\nu} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{a}_{isn}(\omega)$. Finally, this also implies that $\hat{\ell}_{isn}(\omega) = \frac{1}{\eta}\left(\hat{a}_{isn}(\omega) - \hat{w}_{is}(\omega)\right)$, so that labor and wages converge to their competitive counterparts.

Next, we show that profits of firms also converge as $\kappa \to \infty$. Recall that

$$\begin{aligned}
\Psi^b(m) &= -(1+\kappa) w_i^\kappa \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathbb{E}\left[\hat{\Lambda}_{isn}(\omega)(\hat{w}_{isn}(\omega) - \hat{w}_{is}(\omega))\right] ds \\
&= -\int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa(\kappa-\nu)\overline{\ell}_{isn}/\overline{\ell}_{is}}{\kappa + (\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}} \mathbb{E}\left[(\hat{w}_{isn}(\omega) - \hat{w}_{is}(\omega))^2\right] ds
\end{aligned}$$

$$= -\int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \frac{\kappa(\kappa-\nu)\overline{\ell}_{isn}/\overline{\ell}_{is}}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}} \frac{\mathbb{E}\left[(\hat{a}_{isn}(\omega)-(1+\eta\nu)\hat{w}_{is}(\omega))^2\right]}{\left(1+\eta\kappa-\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\frac{\kappa-\nu}{\kappa+(\nu-\kappa)\overline{\ell}_{isn}/\overline{\ell}_{is}}\right)^2} ds,$$

which converges to 0 as $\kappa \to \infty$. Therefore, payments to labor converge to $(1-\eta) z_i m_i^{\eta} \ell_i^{1-\eta} \tilde{\Phi}^b(m)$, and profits converge to the competitive profits. □

B Proofs of Theoretical Results

This Appendix presents the proofs of the theoretical results in Section 2 and the technical lemmas used in those proofs. Throughout the appendix, we will use $\overline{x}$ to denote the value of x with no ex-post shocks, and we will use $\hat{x}$ to denote log deviations from that value. Furthermore, we prove the results for the general case with finite κ and ν. To get the results in Section 2, take the limit as $\kappa \to \infty$ and $\nu \to 0$.

B.1 Proofs of Technical Lemmas

In this section, we prove the Technical Lemmas used below to prove the results in the main text.

Lemma 7. *Average HHI in a location has the following properties:*

(i) $\frac{\partial}{\partial \log m}\left[\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n\in\mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\right)^2 ds\right] < 0$; *and*

(ii) $\frac{\partial}{\partial \log m}\left[\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n\in\mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\right)^2 ds\right] \to 0$ *as* $m \to \infty$.

Proof. We denote the average sectoral HHI by

$$H(m) \equiv \int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n\in\mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\right)^2 ds.$$

Recall that $\overline{\ell}_{isn} = \left(\frac{(1-\eta)z_{isn}}{\overline{w}_i}\right)^{\frac{1}{\eta}}$ where $\overline{w}_i = (1-\eta)\ell_i^{-\eta} m_i^{\eta} z_i$. Therefore,

$$\begin{aligned} H(m) &= \int_0^1 \frac{\ell_i m_i^{-1} z_i^{-\frac{1}{\eta}} \sum_{n\in\mathcal{N}_{is}} z_{isn}^{\frac{1}{\eta}}}{\ell_i} \sum_{n\in\mathcal{N}_{is}} \left(\frac{z_{isn}^{\frac{1}{\eta}}}{\sum_{n'\in\mathcal{N}_{is}} z_{isn'}^{\frac{1}{\eta}}}\right)^2 ds \\ &= \int_0^1 \frac{N_{is}}{m_i} \frac{\sum_{n\in\mathcal{N}_{is}} z_{isn}^{1/\eta}}{N_{is} z_i^{1/\eta}} \frac{\sum_{n\in\mathcal{N}_{is}} (z_{isn}^{1/\eta})^2}{\left(\sum_{n\in\mathcal{N}_{is}} z_{isn}^{1/\eta}\right)^2} ds \end{aligned}$$

We can then decompose this into an expectation over the number of firms in the sector and, conditional on the number of firms, the expected productivity shocks. Then

$$H(m) = \sum_{N=0}^{\infty} \frac{N}{m} \psi_N \frac{m^N e^{-m}}{N!},$$

where

$$\psi_N \equiv \mathbb{E}\left[\frac{\sum_{n\in\mathcal{N}} z_{isn}^{1/\eta}}{N z_i^{1/\eta}} \frac{\sum_{n\in\mathcal{N}} (z_{isn}^{1/\eta})^2}{\left(\sum_{n\in\mathcal{N}} z_{isn}^{1/\eta}\right)^2} \right],$$

for a given N. First note that $\frac{\sum_{n\in\mathcal{N}} (z_{isn}^{1/\eta})^2}{\left(\sum_{n\in\mathcal{N}} z_{isn}^{1/\eta}\right)^2} \in (0,1)$, so

$$\left| \frac{\sum_{n\in\mathcal{N}} z_{isn}^{1/\eta}}{N z_i^{1/\eta}} \frac{\sum_{n\in\mathcal{N}} (z_{isn}^{1/\eta})^2}{\left(\sum_{n\in\mathcal{N}} z_{isn}^{1/\eta}\right)^2} \right| \leq \frac{\sum_{n\in\mathcal{N}} z_{isn}^{1/\eta}}{N z_i^{1/\eta}}.$$

Then as $\mathbb{E}\left[\frac{\sum_{n\in\mathcal{N}_{is}} z_{isn}^{1/\eta}}{N_{is} z_i^{1/\eta}} \right] = 1$, ψ_N exists for each N. We can then rewrite ψ_N,

$$\begin{aligned}
\psi_N &= \frac{1}{N z_i^{1/\eta}} \mathbb{E}\left[\frac{\sum_{n\in\mathcal{N}} (z_{isn}^{1/\eta})^2}{\sum_{n\in\mathcal{N}} z_{isn}^{1/\eta}} \right] \\
&= z_i^{-1/\eta} \mathbb{E}\left[\frac{z_{is1}^{2/\eta}}{\sum_n z_{isn}^{1/\eta}} \right] \\
&= z_i^{-1/\eta} \mathbb{E}\left[\int_0^\infty z_{is1}^{2/\eta} e^{-t\left(\sum_n z_{isn}^{1/\eta}\right)} dt \right],
\end{aligned}$$

where we use the fact that $\frac{1}{x} = \int_0^\infty e^{-tx}$.

By Tonelli's theorem, we switch the orders of integration and use independence to write

$$\psi_N = z_i^{-1/\eta} \int_0^\infty \mathbb{E}[z_{is1}^{2/\eta} e^{-t z_{is1}^{1/\eta}}] \prod_{n\geq 2} \mathbb{E}[e^{-t z_{isn}^{1/\eta}}] dt.$$

We then define

$$g(t) \equiv \mathbb{E}[z_{is1}^{2/\eta} e^{-t z_{is1}^{1/\eta}}]; \quad \phi(t) \equiv \mathbb{E}[e^{-t z_{is2}^{1/\eta}}].$$

We have that $\phi(t)$ exists for all t as $e^{-t z_{is2}^{1/\eta}}$ is bounded above by 1. $g(t)$ exists for all $t > 0$ as $z_{is1}^{2/\eta} e^{-t z_{is1}^{1/\eta}}$ is maximized at $z_{isn}^{1/\eta} = \frac{2}{t}$, and so is bounded above as well. Therefore,

$$\psi_N = z_i^{-1/\eta} \int_0^\infty g(t) \phi(t)^{N-1} dt.$$

We then substitute this into the expression for $H(m)$,

$$H(m) = \sum_{N=0}^{\infty} \frac{N}{m} \psi_N \frac{m^N e^{-m}}{N!}$$

$$
\begin{aligned}
&= \lim_{K\to\infty} \sum_{N=1}^{K} \int_0^\infty g(t)\phi(t)^{N-1} \frac{m^{N-1}e^{-m}}{(N-1)!} dt \\
&= \lim_{K\to\infty} \int_0^\infty g(t) \sum_{k=0}^{K} \phi(t)^k \frac{m^k e^{-m}}{k!} dt \\
&= \int_0^\infty g(t) \sum_{k=0}^{\infty} \phi(t)^k \frac{m^k e^{-m}}{k!} dt \\
&= \int_0^\infty g(t) e^{-m} e^{m\phi(t)} dt = \int_0^\infty g(t) e^{-m(1-\phi(t))} dt,
\end{aligned}
$$

where we use the monotone convergence theorem to swap the order of limits.

We complete the proof by taking the derivative with respect to m. The integrand is bounded above by $g(t)$ for all m so we can bring the derivative under the integral.

$$
\frac{\partial H(m)}{\partial \log m} = mH'(m) = -\int_0^\infty (1-\phi(t)) m g(t) e^{-m(1-\phi(t))} dt < 0,
$$

using the fact that $\phi(t) \in (0,1)$. Next, we take the limit as $m \to \infty$. Note that $m(1-\phi(t))e^{-m(1-\phi(t))} g(t) \leq g(t)$, which is integrable. Thus, we can swap the limits to get,

$$
\begin{aligned}
\lim_{m\to\infty} \frac{\partial H(m)}{\partial \log m} &= -\lim_{m\to\infty} \int_0^\infty (1-\phi(t)) m g(t) e^{-m(1-\phi(t))} dt \\
&= -\int_0^\infty g(t) \lim_{m\to\infty} (1-\phi(t)) m e^{-m(1-\phi(t))} dt = 0.
\end{aligned}
$$

□

Lemma 8. *The regional production function is* $Y_i(\ell, m) = z_i m^\eta \ell^{1-\eta} \tilde{\Phi}(m)$, *where* $z_i \equiv \mathbb{E}[z_{isn}^{1/\eta}]^\eta$ *and* $\tilde{\Phi}(m)$ *is given by,*

$$
\begin{aligned}
\tilde{\Phi}(m) \equiv\ & \mathbb{E}[a_{sn}(\omega)] + (1-\eta) \int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n\in\mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \mathbb{E}[\hat{a}_{sn}(\omega)\hat{\ell}_{sn}(\omega)] ds \\
& - \eta \frac{1-\eta}{2} \int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n\in\mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \mathbb{E}[\hat{\ell}_{sn}(\omega)^2] ds \\
& - \frac{1}{\kappa} \frac{1-\eta}{2} \int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n\in\mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \mathbb{E}\left[\left(\hat{\ell}_{sn}(\omega) - \hat{\ell}_{sn}\right)^2\right] ds \\
& - \left(\frac{1}{\nu} - \frac{1}{\kappa}\right) \frac{1-\eta}{2} \int_0^1 \frac{\overline{\ell}_s}{\ell} \mathbb{E}\left[\left(\hat{\ell}_s(\omega) - \hat{\ell}_s\right)^2\right] ds. \qquad (36)
\end{aligned}
$$

Proof. Expected production is $\int_0^1 \sum_{n\in\mathcal{N}_{is}} z_{isn} a_{isn}(\omega) \ell_{isn}(\omega)^{1-\eta} ds$. We will do a second order approximation around the point $\log \tilde{a}_{isn}(\omega) = \log \tilde{A}_{is}(\omega) = 0$.

We start by characterizing the solution at that point. We find that there is some $\overline{w}_i$ such that

$$
\overline{w}_i = (1-\eta) z_{isn} (\overline{\ell}_{isn})^{-\eta},
$$

so that $\overline{\ell}_{isn} = \left(\frac{(1-\eta)z_{isn}}{\overline{w}_i}\right)^{\frac{1}{\eta}}$. Then labor clearing requires

$$\ell_i = \int_0^1 \ell_i \overline{L}_{is} ds = \int_0^1 \sum_{n \in \mathcal{N}_{is}} \overline{\ell}_{isn} ds = \int_0^1 \sum_{n \in \mathcal{N}_{is}} \left(\frac{(1-\eta)z_{isn}}{\overline{w}_i}\right)^{\frac{1}{\eta}} ds.$$

This implies that $\overline{w}_i = (1-\eta)\ell_i^{-\eta} m_i^{\eta} \mathbb{E}\left[z_{isn}^{1/\eta}\right]^{\eta}$. Then production is

$$\overline{Y}_i = \int_0^1 \sum_{n \in \mathcal{N}_{is}} z_{isn} \left(\frac{(1-\eta)z_{isn}}{\overline{w}_i}\right)^{\frac{1-\eta}{\eta}} ds = z_i \ell_i^{1-\eta} m_i^{\eta},$$

where $z_i = \mathbb{E}\left[z_{isn}^{1/\eta}\right]^{\eta}$.

Taking the log second-order approximation to production gives

$$\begin{aligned} Y_i \approx z_i (m_i)^{\eta} (\ell_i)^{1-\eta} \int_0^1 \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_i} \Bigg(1 + \left(\hat{a}_{isn}(\omega) + (1-\eta)\hat{\ell}_{isn}(\omega)\right) \\ + \frac{1}{2}\left(\hat{a}_{isn}(\omega) + (1-\eta)\hat{\ell}_{isn}(\omega)\right)^2 \Bigg) ds. \end{aligned}$$

To transform this to be completely second order, we do a second-order approximation to the labor constraints,

$$\begin{aligned} -\frac{1}{\kappa}\hat{\ell}_{is} + \frac{1+\kappa}{\kappa}\hat{\ell}_{is}(\omega) + \frac{1}{2}\left(-\frac{1}{\kappa}\hat{\ell}_{is} + \frac{1+\kappa}{\kappa}\hat{\ell}_{is}(\omega)\right)^2 = \\ \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \left[-\frac{1}{\kappa}\hat{\ell}_{isn} + \frac{1+\kappa}{\kappa}\hat{\ell}_{isn}(\omega) + \frac{1}{2}\left(-\frac{1}{\kappa}\hat{\ell}_{isn} + \frac{1+\kappa}{\kappa}\hat{\ell}_{isn}(\omega)\right)^2 \right], \\ 0 = \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \left[-\frac{1}{\nu}\hat{\ell}_{is} + \frac{1+\nu}{\nu}\hat{\ell}_{is}(\omega) + \frac{1}{2}\left(-\frac{1}{\nu}\hat{\ell}_{is} + \frac{1+\nu}{\nu}\hat{\ell}_{is}(\omega)\right)^2 \right] ds, \\ \hat{\ell}_{is} + \frac{1}{2}\hat{\ell}_{is}^2 = \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \left(\hat{\ell}_{isn} + \frac{1}{2}\hat{\ell}_{isn}^2\right), \\ \text{and} \qquad 0 = \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \left(\hat{\ell}_{is} + \frac{1}{2}\hat{\ell}_{is}^2\right) ds, \end{aligned}$$

where we use the fact $\hat{\ell}_{isn}(\omega) = \hat{L}_{isn}(\omega)$.

Then we can transform $\int_0^1 \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_i} \hat{\ell}_{isn}(\omega) ds$ to second order. That is,

$$\begin{aligned} \int_0^1 \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_i} \hat{\ell}_{isn}(\omega) ds = -\frac{1}{2}\frac{\kappa}{1+\kappa} \int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \left(-\frac{1}{\kappa}\hat{\ell}_{isn} + \frac{1+\kappa}{\kappa}\hat{\ell}_{isn}(\omega)\right)^2 ds \\ + \frac{1}{2}\frac{\kappa}{1+\kappa} \int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \left(-\frac{1}{\kappa}\hat{\ell}_{is} + \frac{1+\kappa}{\kappa}\hat{\ell}_{is}(\omega)\right)^2 ds \end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{1+\kappa}\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{\ell}_{isn}ds-\frac{1}{1+\kappa}\int_0^1\frac{\overline{\ell}_{is}}{\ell_i}\hat{\ell}_{is}ds+\int_0^1\frac{\overline{\ell}_{is}}{\ell_i}\hat{\ell}_{is}(\omega)ds\\
&=-\frac{1}{2}\frac{\kappa}{1+\kappa}\int_0^1\frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\left(-\frac{1}{\kappa}\hat{\ell}_{isn}+\frac{1+\kappa}{\kappa}\hat{\ell}_{isn}(\omega)\right)^2ds\\
&+\frac{1}{2}\frac{\kappa}{1+\kappa}\int_0^1\frac{\overline{\ell}_{is}}{\overline{\ell}_i}\left(-\frac{1}{\kappa}\hat{\ell}_{is}+\frac{1+\kappa}{\kappa}\hat{\ell}_{is}(\omega)\right)^2ds\\
&-\frac{1}{2}\frac{\nu}{1+\nu}\int_0^1\frac{\overline{\ell}_{is}}{\ell_i}\left(-\frac{1}{\nu}\hat{\ell}_{is}+\frac{1+\nu}{\nu}\hat{\ell}_{is}(\omega)\right)^2ds+\frac{1}{1+\nu}\int_0^1\frac{\overline{\ell}_{is}}{\ell_i}\hat{\ell}_{is}ds\\
&+\frac{1}{1+\kappa}\int_0^1\frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{\ell}_{isn}ds-\frac{1}{1+\kappa}\int_0^1\frac{\overline{\ell}_{is}}{\ell_i}\hat{\ell}_{is}ds\\
&=-\frac{1}{2}\frac{\kappa}{1+\kappa}\int_0^1\frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\left(-\frac{1}{\kappa}\hat{\ell}_{isn}+\frac{1+\kappa}{\kappa}\hat{\ell}_{isn}(\omega)\right)^2ds\\
&+\frac{1}{2}\frac{\kappa}{1+\kappa}\int_0^1\frac{\overline{\ell}_{is}}{\overline{\ell}_i}\left(-\frac{1}{\kappa}\hat{\ell}_{is}+\frac{1+\kappa}{\kappa}\hat{\ell}_{is}(\omega)\right)^2ds\\
&-\frac{1}{2}\frac{\nu}{1+\nu}\int_0^1\frac{\overline{\ell}_{is}}{\ell_i}\left(-\frac{1}{\nu}\hat{\ell}_{is}+\frac{1+\nu}{\nu}\hat{\ell}_{is}(\omega)\right)^2ds\\
&-\frac{1}{2}\frac{1}{1+\kappa}\int_0^1\frac{\overline{\ell}_{is}}{\ell_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{\ell}_{isn}^2ds\\
&+\frac{1}{2}\frac{1}{1+\kappa}\int_0^1\frac{\overline{\ell}_{is}}{\ell_i}\hat{\ell}_{is}^2ds-\frac{1}{2}\frac{1}{1+\nu}\int_0^1\frac{\overline{\ell}_{is}}{\overline{\ell}_i}\hat{\ell}_{is}^2ds\\
&=-\frac{1}{2}\int_0^1\frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{\ell}_{isn}(\omega)^2ds\\
&-\frac{1}{2}\frac{1}{\kappa}\int_0^1\frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\left(\hat{\ell}_{isn}(\omega)-\hat{\ell}_{isn}\right)^2ds\\
&-\frac{1}{2}\left(\frac{1}{\nu}-\frac{1}{\kappa}\right)\int_0^1\frac{\overline{\ell}_{is}}{\overline{\ell}_i}\left(\hat{\ell}_{is}(\omega)-\hat{\ell}_{is}\right)^2ds.
\end{aligned}
$$

Substituting this into the expression for the log second-order approximation to production gives

$$
\begin{aligned}
\frac{Y_i}{z_i(m_i)^\eta(\ell_i)^{1-\eta}}\approx 1&+\mathbb{E}[\hat{a}_{isn}(\omega)]+\frac{1}{2}\mathbb{E}[\hat{a}_{isn}(\omega)^2]\\
&+(1-\eta)\int_0^1\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\ell_i}\hat{a}_{isn}(\omega)\hat{\ell}_{isn}(\omega)ds\\
&+\frac{(1-\eta)^2}{2}\int_0^1\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\ell_i}\hat{\ell}_{isn}(\omega)^2ds\\
&-\frac{1-\eta}{2}\int_0^1\frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{\ell}_{isn}(\omega)^2ds
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{1-\eta}{2}\frac{1}{\kappa}\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\left(\hat{\ell}_{isn}(\omega)-\hat{\ell}_{isn}\right)^2 ds \\
&- \frac{1-\eta}{2}\left(\frac{1}{\nu}-\frac{1}{\kappa}\right)\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i}\left(\hat{\ell}_{is}(\omega)-\hat{\ell}_{is}\right)^2 ds \\
\approx\ & \mathbb{E}[a_{isn}(\omega)] + (1-\eta)\int_0^1 \sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\ell_i}\hat{a}_{isn}(\omega)\hat{\ell}_{isn}(\omega)ds \\
&- \frac{1-\eta}{2}\eta\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{\ell}_{isn}(\omega)^2 ds \\
&- \frac{1-\eta}{2}\frac{1}{\kappa}\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i}\sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\left(\hat{\ell}_{isn}(\omega)-\hat{\ell}_{isn}\right)^2 ds \\
&- \frac{1-\eta}{2}\left(\frac{1}{\nu}-\frac{1}{\kappa}\right)\int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i}\left(\hat{\ell}_{is}(\omega)-\hat{\ell}_{is}\right)^2 ds,
\end{aligned}
$$

where we use the fact that to log second order, $\mathbb{E}[a_{sn}(\omega)] \approx 1 + \mathbb{E}[\hat{a}_{sn}(\omega)] + \frac{1}{2}\mathbb{E}[\hat{a}_{sn}(\omega)^2]$. □

B.2 Proofs of Propositions & Lemmas in Main Text

In this subsection, we prove the Propositions and Lemmas in the main text.

Proof of Lemma 1. By Lemma 8, the regional production can be written $Y_i(\ell, m) = z_i m^{\eta}\ell^{1-\eta}\tilde{\Phi}(m)$ where

$$
\begin{aligned}
\tilde{\Phi}(m) \equiv\ & \mathbb{E}[a_{sn}(\omega)] + (1-\eta)\int_0^1 \frac{\overline{\ell}_s}{\ell}\sum_{n\in\mathcal{N}_s}\frac{\overline{\ell}_{sn}}{\overline{\ell}_s}\mathbb{E}[\hat{a}_{sn}(\omega)\hat{\ell}_{sn}(\omega)]ds \\
&- \eta\frac{1-\eta}{2}\int_0^1 \frac{\overline{\ell}_s}{\ell}\sum_{n\in\mathcal{N}_s}\frac{\overline{\ell}_{sn}}{\overline{\ell}_s}\mathbb{E}[\hat{\ell}_{sn}(\omega)^2]ds \\
&- \frac{1}{\kappa}\frac{1-\eta}{2}\int_0^1 \frac{\overline{\ell}_s}{\ell}\sum_{n\in\mathcal{N}_s}\frac{\overline{\ell}_{sn}}{\overline{\ell}_s}\mathbb{E}\left[\left(\hat{\ell}_{sn}(\omega)-\hat{\ell}_{sn}\right)^2\right] ds \\
&- \left(\frac{1}{\nu}-\frac{1}{\kappa}\right)\frac{1-\eta}{2}\int_0^1 \frac{\overline{\ell}_s}{\ell}\mathbb{E}\left[\left(\hat{\ell}_s(\omega)-\hat{\ell}_s\right)^2\right] ds
\end{aligned}
$$

Therefore, the planner is looking to maximize this second-order production function subject to the labor constraints. Following Benigno and Woodford (2003) and Benigno and Woodford (2012), we can do a linear approximation to the labor constraints, embedded in the sets $\mathcal{L}$ and $\mathcal{L}_\Omega(\cdot)$,

$$
\begin{aligned}
\hat{\ell}_{is}(\omega) &= \sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{\ell}_{isn}(\omega), & 0 &= \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i}\hat{\ell}_{is}(\omega)ds, \\
\hat{\ell}_{is} &= \sum_{n\in\mathcal{N}_{is}}\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{\ell}_{isn}, & 0 &= \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i}\hat{\ell}_{is}ds.
\end{aligned}
$$

To simplify the explication, we rewrite the maximization problem with vector notation as

$$\begin{aligned} \max_{x} \quad & ax - \frac{1}{2}x'bx \\ \text{s.t.} \quad & cx = 0 \end{aligned}$$

where x is the vector of labor supply decisions, a is the vector of productivity shocks so that

$$ax = (1-\eta)\int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n\in\mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \mathbb{E}[\hat{a}_{sn}(\omega)\hat{\ell}_{sn}(\omega)]ds,$$

b is the self-adjoint operator (i.e., symmetric matrix) representing the loss function so that

$$\begin{aligned} x'bx = \eta(1-\eta)\int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n\in\mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \mathbb{E}[\hat{\ell}_{sn}(\omega)^2]ds \\ + \frac{1}{\kappa}(1-\eta)\int_0^1 \frac{\overline{\ell}_s}{\ell} \sum_{n\in\mathcal{N}_s} \frac{\overline{\ell}_{sn}}{\overline{\ell}_s} \mathbb{E}\left[\left(\hat{\ell}_{sn}(\omega) - \hat{\ell}_{sn}\right)^2\right] ds \\ + \left(\frac{1}{\nu} - \frac{1}{\kappa}\right)(1-\eta)\int_0^1 \frac{\overline{\ell}_s}{\ell} \mathbb{E}\left[\left(\hat{\ell}_s(\omega) - \hat{\ell}_s\right)^2\right] ds, \end{aligned}$$

and c is the matrix representing the linear constraints. Forming the Lagrangian, we have

$$ax - \frac{1}{2}x'bx - \lambda cx.$$

Taking the FOCs, we get

$$a - (bx)' - \lambda c = 0.$$

Therefore, $x = b^{-1}(a - \lambda c)' = b^{-1}(a' - c'\lambda')$. Using the constraint, we can solve for λ:

$$\begin{aligned} 0 = cx &= cb^{-1}(a' - c'\lambda') \\ \lambda' &= (cb^{-1}c')^{-1}cb^{-1}a'. \end{aligned}$$

Then, to complete the proof, we will show that $ax = x'bx$.

$$\begin{aligned} x'bx &= \left(b^{-1}(a' - c'(cb^{-1}c')^{-1}cb^{-1}a')\right)' bb^{-1}(a' - c'(cb^{-1}c')^{-1}cb^{-1}a') \\ &= (a - ab^{-1}c'(cb^{-1}c')^{-1}c)b^{-1}(a' - c'(cb^{-1}c')^{-1}cb^{-1}a') \\ &= ab^{-1}a' - ab^{-1}c'(cb^{-1}c')^{-1}cb^{-1}a' - ab^{-1}c'(cb^{-1}c')^{-1}cb^{-1}a' \\ &\quad + ab^{-1}c'(cb^{-1}c')^{-1}cb^{-1}c'(cb^{-1}c')^{-1}cb^{-1}a' \\ &= ab^{-1}a' - ab^{-1}c'(cb^{-1}c')^{-1}cb^{-1}a', \end{aligned}$$

where we use the fact that b and $cb^{-1}c'$ are self adjoint and therefore b^{-1} and $(cb^{-1}c')^{-1}$ are self

adjoint. Similarly,

$$\begin{aligned} ax &= ab^{-1}(a' - c'(cb^{-1}c')^{-1}cb^{-1}a') \\ &= x'bx. \end{aligned}$$

Therefore, $ax - \frac{1}{2}x'bx = \frac{1}{2}ax$, completing the proof. □

Proof of Proposition 1. This result follows almost immediately from Lemma 1. Implicitly differentiating the free entry condition (10) implies that

$$\frac{d\log m_i}{d\log \ell_i} = \frac{1}{1 - \frac{1}{1-\eta}\frac{\partial \log \Phi(m_i)}{\partial \log m_i}}.$$

If profits are decreasing in the number of firms, the denominator must be positive. Therefore, differentiating the expression for wages gives

$$\begin{aligned} \frac{d\log w_i}{d\log \ell_i} &= \left(\eta + \frac{\partial \log \Phi(m_i)}{\partial \log m_i}\right)\frac{d\log m_i}{d\log \ell_i} - \eta \\ &= \frac{\frac{1}{1-\eta}\frac{\partial \log \Phi(m_i)}{\partial \log m_i}}{1 - \frac{1}{1-\eta}\frac{\partial \log \Phi(m_i)}{\partial \log m_i}}, \end{aligned}$$

which is greater than zero if and only if $\frac{\partial \log \Phi(m_i)}{\partial \log m_i} > 0$. □

Proof of Proposition 2. We prove the result for finite κ and ν where labor is more substitutable across firms within a sector than across sectors, i.e., $\kappa > \nu$. Taking a log first order approximation to the labor supply FOCs (27) after substituting out $\lambda_{is}(\omega)$ and λ_{is} implies,

$$\begin{aligned} \hat{w}_{isn}(\omega) &= \hat{\lambda}(\omega) + \left(\frac{1}{\nu} - \frac{1}{\kappa}\right)\left(\hat{\ell}_{is}(\omega) - \hat{\ell}_{is}\right) + \frac{1}{\kappa}\left(\hat{\ell}_{isn}(\omega) - \hat{\ell}_{isn}\right) \\ \hat{\bar{\lambda}}_i &= \frac{1+\nu}{1+\kappa}\mathbb{E}\left[\hat{w}_{isn}(\omega)\right] + \frac{\kappa-\nu}{1+\kappa}\mathbb{E}\left[\hat{\lambda}(\omega)\right] + \frac{\kappa}{1+\kappa}\frac{1+\nu}{\nu}\mathbb{E}\left[\hat{\ell}_{isn}(\omega) - \hat{\ell}_{isn}\right], \end{aligned}$$

where we use the fact that $\hat{L}_{isn}(\omega) = \hat{\ell}_{isn}(\omega)$.

Taking a log first-order approximation to the labor constraints embedded in $\mathcal{L}$ and $\mathcal{L}_\Omega(\cdot)$ implies,

$$\begin{aligned} \hat{\ell}_{is}(\omega) &= \sum_{n\in\mathcal{N}_{is}} \frac{\bar{\ell}_{isn}}{\bar{\ell}_{is}}\hat{\ell}_{isn}(\omega), & 0 &= \int_0^1 \frac{\bar{\ell}_{is}}{\ell_i}\hat{\ell}_{is}(\omega)ds, \\ \hat{\ell}_{is} &= \sum_{n\in\mathcal{N}_{is}} \frac{\bar{\ell}_{isn}}{\bar{\ell}_{is}}\hat{\ell}_{isn}, & 0 &= \int_0^1 \frac{\bar{\ell}_{is}}{\ell_i}\hat{\ell}_{is}ds. \end{aligned}$$

And the labor demand curve implies that $\hat{a}_{isn}(\omega) - \eta\hat{\ell}_{isn}(\omega) = \hat{w}_{isn}(\omega)$. First note that taking expectations yields $\mathbb{E}[\hat{a}_{isn}(\omega)] - \eta\mathbb{E}[\hat{\ell}_{isn}(\omega)] = \mathbb{E}[\hat{w}_{isn}(\omega)]$, which implies $\mathbb{E}[\hat{w}_{isn}(\omega)] = -\eta\mathbb{E}[\hat{\ell}_{isn}(\omega)]$.

Therefore,

$$-\hat{\lambda}_i(\omega) = \left(\frac{1}{\nu} - \frac{1}{\kappa}\right)\left(\hat{\ell}_{is}(\omega) - \hat{\ell}_{is}\right) + \frac{1}{\kappa}\left(\hat{\ell}_{isn}(\omega) - \hat{\ell}_{isn}\right) + \eta\hat{\ell}_{isn}(\omega)$$

$$-\int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{\lambda}(\omega)ds = \left(\frac{1}{\nu} - \frac{1}{\kappa}\right)\int_0^1 \frac{\overline{\ell}_{is}}{\ell_i}\left(\hat{\ell}_{is}(\omega) - \hat{\ell}_{is}\right)ds$$

$$+ \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\left(\frac{1}{\kappa}\left(\hat{\ell}_{isn}(\omega) - \hat{\ell}_{isn}\right) + \eta\hat{\ell}_{isn}(\omega)\right)ds$$

$$\hat{\lambda}(\omega) = 0.$$

Then

$$\hat{\lambda}_i = \frac{1+\nu}{1+\kappa}\mathbb{E}\left[\hat{w}_{isn}(\omega)\right] + \frac{\kappa-\nu}{1+\kappa}\mathbb{E}\left[\hat{\lambda}(\omega)\right] + \frac{\kappa}{1+\kappa}\frac{1+\nu}{\nu}\mathbb{E}\left[\hat{\ell}_{isn}(\omega) - \hat{\ell}_{isn}\right]$$

$$\hat{\lambda}_i = \frac{1+\nu}{1+\kappa}\left(\frac{\kappa}{\nu} - \eta\right)\mathbb{E}\left[\hat{\ell}_{isn}(\omega)\right] - \frac{\kappa}{1+\kappa}\frac{1+\nu}{\nu}\hat{\ell}_{isn}.$$

Again, taking a weighted sum across all firms implies that $\hat{\lambda}_i = 0$. Since $\kappa/\nu > 1$ and $\eta \in (0,1)$, the right-hand side is a non-degenerate linear system of equations equal to 0. It then follows that $\hat{\ell}_{isn} = \mathbb{E}[\hat{\ell}_{isn}(\omega)] = \mathbb{E}[\hat{w}_{isn}(\omega)] = 0$ because otherwise it is not possible for $0 = \left(\frac{\kappa}{\nu} - \eta\right)\mathbb{E}\left[\hat{\ell}_{isn}(\omega)\right] - \frac{\kappa}{\nu}\hat{\ell}_{isn}$ and the labor constraints to hold.

Therefore, we can solve for sectoral labor

$$\hat{a}_{isn}(\omega) - \eta\hat{\ell}_{isn}(\omega) = \left(\frac{1}{\nu} - \frac{1}{\kappa}\right)\hat{\ell}_{is}(\omega) + \frac{1}{\kappa}\hat{\ell}_{isn}(\omega)$$

$$\sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{a}_{isn}(\omega) = \left(\frac{1}{\nu} - \frac{1}{\kappa}\right)\hat{\ell}_{is}(\omega) + \left(\frac{1}{\kappa} + \eta\right)\hat{\ell}_{is}(\omega)$$

$$\frac{1}{\eta + \frac{1}{\nu}} \sum_{n\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{a}_{isn}(\omega) = \hat{\ell}_{is}(\omega).$$

And individual firm labor is

$$\hat{\ell}_{isn}(\omega) = \frac{1}{\eta + \frac{1}{\kappa}}\left(\hat{a}_{isn}(\omega) - \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}} \sum_{n'\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}}\hat{a}_{isn'}(\omega)\right).$$

Therefore,

$$\mathbb{E}[\hat{a}_{isn}(\omega)\hat{\ell}_{isn}(\omega)] = \mathbb{E}\left[\hat{a}_{isn}(\omega)\frac{1}{\eta + \frac{1}{\kappa}}\left(\hat{a}_{isn}(\omega) - \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}} \sum_{n'\in\mathcal{N}_{is}} \frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}}\hat{a}_{isn'}(\omega)\right)\right]$$

$$= \frac{1}{\eta + \frac{1}{\kappa}}\left(\sigma_S^2 + \sigma_N^2\right) - \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\left(\eta + \frac{1}{\kappa}\right)\left(\eta + \frac{1}{\nu}\right)}\left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\sigma_N^2 + \sigma_S^2\right)$$

$$= \frac{1}{\eta + \frac{1}{\nu}} \sigma_S^2 + \frac{\eta + \frac{1}{\nu} - \left(\frac{1}{\nu} - \frac{1}{\kappa}\right) \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}}{\left(\eta + \frac{1}{\kappa}\right)\left(\eta + \frac{1}{\nu}\right)} \sigma_N^2.$$

Then we can calculate $\Phi(m)$,

$$\begin{aligned}
\Phi(m) &= \mathbb{E}[a_{isn}(\omega)] + \frac{1-\eta}{2} \int_0^1 \frac{\overline{\ell}_{is}}{\overline{\ell}_i} \sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathbb{E}\left[\hat{a}_{isn}(\omega) \hat{\ell}_{isn}(\omega)\right] ds \\
&= \mathbb{E}[a_{isn}(\omega)] + \frac{1-\eta}{2} \frac{1}{\eta + \frac{1}{\nu}} \sigma_S^2 + \frac{1-\eta}{2} \frac{\eta + \frac{1}{\nu} - \left(\frac{1}{\nu} - \frac{1}{\kappa}\right) \int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_{n \in \mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\right)^2 ds}{\left(\eta + \frac{1}{\kappa}\right)\left(\eta + \frac{1}{\nu}\right)} \sigma_N^2.
\end{aligned}$$

Therefore,

$$\frac{\partial \log \Phi(m)}{\partial \log m} = -\frac{1-\eta}{2} \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\left(\eta + \frac{1}{\kappa}\right)\left(\eta + \frac{1}{\nu}\right)} \frac{\partial}{\partial \log m} \left[\int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_{n \in \mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\right)^2 ds\right] \frac{\sigma_N^2}{\Phi(m)}$$

By Lemma 7, $\frac{\partial}{\partial \log m}\left[\int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_{n \in \mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\right)^2 ds\right] < 0$. Then by Proposition 1, the first result will follow. Similarly, by Lemma 7, $\frac{\partial}{\partial \log m}\left[\int_0^1 \frac{\overline{\ell}_{is}}{\ell_i} \sum_{n \in \mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\right)^2 ds\right] \to 0$ as $m \to \infty$ so

$$\frac{d \log w_i}{d \log \ell_i} = \frac{\frac{1}{1-\eta} \frac{\partial \log \Phi(m_i)}{\partial \log m_i}}{1 - \frac{1}{1-\eta} \frac{\partial \log \Phi(m_i)}{\partial \log m_i}} \to 0,$$

as $m \to \infty$. Noting that $m \to \infty$ as $\ell \to \infty$ completes the proof. □

Proof of Proposition 3. First-best entry must satisfy,

$$\psi_i = \left(1 + \frac{1}{\eta} \frac{\partial \log \Phi(m_i^{FB})}{\partial \log m}\right) \frac{\eta Y_i}{m_i^{FB}}.$$

In equilibrium, entry satisfies

$$\psi_i = (1 + \tau_i) \frac{\eta Y_i}{m_i},$$

where τ_i is the subsidy on entry. Thus, in any optimal equilibrium, $\tau_i = \frac{1}{\eta} \frac{\partial \log \Phi(m_i^{FB})}{\partial \log m}$. Furthermore, $\frac{\partial \log \Phi(m)}{\partial \log m} \to 0$ as $\ell \to \infty$ as we showed in Proposition 2. □

Proof of Lemma 2. This result follows immediately from Lemma 8 and the fact that, because perfect competition is efficient, it must produce the most conditional on the number of firms and workers. □

Proof of Lemma 3. This result follows for Cournot and Bertrand competition from Lemmas 4 and 5 and the fact that wages must have a markdown. □

Proof of Proposition 4. At the first best, workers need to earn their marginal product in each location. That is,

$$(1+\tau_i^w)w_i^{FB} = \frac{\partial Y_i(\ell_i^{FB}, m_i^{FB})}{\partial \ell} = \frac{(1-\eta)Y_i^{FB}}{\ell_i^{FB}}.$$

Furthermore, firms need to be paid their marginal product,

$$\psi_i = \frac{\partial Y_i(\ell_i^{FB}, m_i^{FB})}{\partial m} = \left(1 + \frac{1}{\eta}\frac{\partial \log \Phi^f(m_i^{FB})}{\partial \log m}\right)\frac{\eta Y_i^{FB}}{m_i^{FB}}.$$

In equilibrium, workers are paid

$$\begin{aligned} w_i &= \frac{(1-\eta)z_i(m_i)^\eta(\ell_i)^{1-\eta}(\Phi^f(m_i)+\Psi^f(m_i))}{\ell_i} \\ &= \left(1 + \frac{\Psi^f(m_i)}{\Phi^f(m_i)}\right)\frac{(1-\eta)Y_i}{\ell_i}. \end{aligned}$$

Meanwhile, free entry implies

$$\begin{aligned} \psi_i &= (1+\tau_i^\pi)\frac{Y_i - w_i\ell_i}{m_i} \\ &= (1+\tau_i^\pi)\frac{z_i(m_i)^\eta(\ell_i)^{1-\eta}\Phi^f(m_i) - (1-\eta)z_i(m_i)^\eta(\ell_i)^{1-\eta}\left(\Phi^f(m_i)+\Psi^f(m_i)\right)}{m_i} \\ &= (1+\tau_i^\pi)\frac{\eta Y_i}{m_i}\left(1 - \frac{1-\eta}{\eta}\frac{\Psi^f(m_i)}{\Phi^f(m_i)}\right). \end{aligned}$$

Equating the equilibrium conditions to the first best optimality conditions implies the results. □

Proof of Proposition 5. Workers are paid their marginal product of labor. Thus, to log first order,

$$\hat{w}_{isn}(\omega) = \hat{a}_{isn}(\omega) - \eta\hat{\ell}_{isn}(\omega).$$

Then, to first order, the short labor supply elasticity across firms is

$$\hat{\ell}_{isn}(\omega) = \kappa\hat{w}_{isn}(\omega) - (\kappa-\nu)\hat{w}_{is}(\omega),$$

where

$$\hat{w}_{is}(\omega) = \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{w}_{isn}(\omega).$$

Summing across labor supply equations, weighted by $\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}$ implies $\sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{\ell}_{isn}(\omega) = \nu\hat{w}_{is}(\omega)$. Therefore,

$$\hat{w}_{is}(\omega) = \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\hat{w}_{isn}(\omega)$$

$$
\begin{aligned}
&= \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{a}_{isn}(\omega) - \eta \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{\ell}_{isn}(\omega) \\
&= \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{a}_{isn}(\omega) - \eta\nu \hat{w}_{is}(\omega) \\
&= \frac{1}{1+\eta\nu} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{a}_{isn}(\omega).
\end{aligned}
$$

Then

$$
\begin{aligned}
\hat{\ell}_{isn}(\omega) &= \kappa \hat{w}_{isn}(\omega) - (\kappa - \nu) \hat{w}_{is}(\omega) \\
&= \kappa \left(\hat{a}_{isn}(\omega) - \eta \hat{\ell}_{isn}(\omega) \right) - \frac{\kappa - \nu}{1+\eta\nu} \sum_{n'} \frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}} \hat{a}_{isn'}(\omega) \\
&= \frac{1}{\eta + \frac{1}{\kappa}} \left(\hat{a}_{isn}(\omega) - \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}} \sum_{n'} \frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}} \hat{a}_{isn'}(\omega) \right).
\end{aligned}
$$

With this, we turn to find the log wage bill.

$$
\begin{aligned}
\log \left(\sum_{n \in \mathcal{N}_{is}} w_{isn}(\omega) \ell_{isn}(\omega) \right) &\approx \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \left(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega) \right) \\
&= \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \left(\hat{a}_{isn}(\omega) + (1-\eta) \hat{\ell}_{isn}(\omega) \right) \\
&= \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \left(\frac{1 + \frac{1}{\kappa}}{\eta + \frac{1}{\kappa}} \hat{a}_{isn}(\omega) - \frac{\left(\frac{1}{\nu} - \frac{1}{\kappa}\right)(1-\eta)}{\left(\eta + \frac{1}{\kappa}\right)\left(\eta + \frac{1}{\nu}\right)} \sum_{n'} \frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}} \hat{a}_{isn'}(\omega) \right) \\
&= \frac{\eta\left(1 + \frac{1}{\kappa}\right) + \frac{1}{\nu}\left(1 + \frac{1}{\kappa}\right) - \frac{1}{\nu}(1-\eta) + \frac{1}{\kappa}(1-\eta)}{\left(\eta + \frac{1}{\kappa}\right)\left(\eta + \frac{1}{\nu}\right)} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{a}_{isn}(\omega) \\
&= \frac{\eta + \frac{1}{\nu}\frac{1}{\kappa} + \eta\frac{1}{\nu} + \frac{1}{\kappa}}{\left(\eta + \frac{1}{\kappa}\right)\left(\eta + \frac{1}{\nu}\right)} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{a}_{isn}(\omega) \\
&= \frac{1 + \frac{1}{\nu}}{\eta + \frac{1}{\nu}} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{a}_{isn}(\omega) = \frac{1+\nu}{1+\eta\nu} \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{a}_{isn}(\omega).
\end{aligned}
$$

We then take the variance. We have,

$$
\begin{aligned}
\text{Var} \left(\sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \left(\hat{w}_{isn}(\omega) + \hat{\ell}_{isn}(\omega) \right) \right) &= \left(\frac{1+\nu}{1+\eta\nu} \right)^2 \text{Var} \left(\sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{a}_{isn}(\omega) \right) \\
&= \left(\frac{1+\nu}{1+\eta\nu} \right)^2 \text{Var} \left(\hat{\bar{A}}_{is}(\omega) + \sum_n \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \hat{\bar{a}}_{isn}(\omega) \right) \\
&= \left(\frac{1+\nu}{1+\eta\nu} \right)^2 \left(\sigma_S^2 + \sum_{n \in \mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \right)^2 \sigma_N^2 \right).
\end{aligned}
$$

To complete the proof, we need to prove that expected HHI is decreasing in the number of firms.

Recall that $\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} = \frac{z_{isn}^{\frac{1}{\eta}}}{\sum_{n'} z_{isn'}^{\frac{1}{\eta}}}$. Denote $x_{isn} \equiv z_{isn}^{\frac{1}{\eta}}$. The x_{isn} are independent Pareto random variables. Then we will prove that

$$\mathbb{E}\left[\frac{\sum_{n=1}^{N+1} x_{isn}^2}{\left(\sum_{n=1}^{N+1} x_{isn}\right)^2}\right] \leq \mathbb{E}\left[\frac{\sum_{n=1}^{N} x_{isn}^2}{\left(\sum_{n=1}^{N} x_{isn}\right)^2}\right].$$

To prove this, consider some arbitrary m. Then we will find the expected HHI with $N+1$ firms when firm n' is the least productive firm and $x_{isn'} = m$. Note that

$$\begin{aligned}
&\mathbb{E}\left[\frac{\sum_{n=1}^{N+1} x_{isn}^2}{\left(\sum_{n=1}^{N+1} x_{isn}\right)^2} | m = x_{n'}, x_n \geq m, \forall n \neq n'\right] \\
&\quad = \mathbb{E}\left[\frac{m^2 + \sum_{n \neq n'} x_{isn}^2}{\left(m + \sum_{n \neq n'} x_{isn}\right)^2} | m = x_{n'}, x_n \geq m, \forall n \neq n'\right] \\
&\quad = \mathbb{E}\left[\frac{1 + \sum_{n \neq n'} \left(\frac{x_{isn}}{m}\right)^2}{\left(1 + \sum_{n \neq n'} \frac{x_{isn}}{m}\right)^2} | m = x_{n'}, \frac{x_n}{m} \geq 1, \forall n \neq n'\right].
\end{aligned}$$

Then since x_{isn} is distributed Pareto, $\frac{x_{isn}}{m}$ is distributed Pareto. Furthermore, since $\frac{x_n}{m}$ is distributed Pareto, $\frac{x_n}{m}$ given it is greater than 1 is distributed Pareto with location parameter 1. Therefore,

$$\mathbb{E}\left[\frac{1 + \sum_{n \neq n'} \left(\frac{x_{isn}}{m}\right)^2}{\left(1 + \sum_{n \neq n'} \frac{x_{isn}}{m}\right)^2} | m = x_{n'}, \frac{x_n}{m} \geq 1, \forall n \neq n'\right] = \mathbb{E}\left[\frac{1 + \sum_{n=1}^{N} y_n^2}{\left(1 + \sum_{n=1}^{N} y_n\right)^2}\right]$$

for iid Pareto variables y_n with location parameter 1. Next, define $A \equiv \sum_{n=1}^{N} y_n^2$ and $B \equiv \sum_{n=1}^{N} y_n$. Note that as $y_n \geq 1$, $A \geq B$. We will prove that

$$\frac{1 + A}{(1 + B)^2} \leq \frac{A}{B^2}.$$

Note,

$$\begin{aligned}
\frac{1 + A}{(1 + B)^2} - \frac{A}{B^2} &= \frac{B^2 + AB^2 - A - 2AB - AB^2}{B^2(1 + B)^2} \\
&= \frac{B^2 - A - 2AB}{B^2(1 + B)^2} < 0.
\end{aligned}$$

Therefore,

$$\mathbb{E}\left[\frac{1 + \sum_{n=1}^{N} y_n^2}{\left(1 + \sum_{n=1}^{N} y_n\right)^2}\right] < \mathbb{E}\left[\frac{\sum_{n=1}^{N} y_n^2}{\left(\sum_{n=1}^{N} y_n\right)^2}\right],$$

and the right hand side is exactly the expected HHI of sectors with N firms as the location parameter does not affect HHI. Putting everything together,

$$\mathbb{E}\left[\frac{\sum_{n=1}^{N+1} x_{isn}^2}{\left(\sum_{n=1}^{N+1} x_{isn}\right)^2} | m = x_{n'}, x_n \geq m, \forall n \neq n'\right] < \mathbb{E}\left[\frac{\sum_{n=1}^{N} y_n^2}{\left(\sum_{n=1}^{N} y_n\right)^2}\right].$$

Taking expectations of both sides proves the result.

□

Proof of Proposition 6. From the previous proof,

$$\hat{\ell}_{isn}(\omega) = \frac{1}{\eta + \frac{1}{\kappa}}\left(\hat{a}_{isn}(\omega) - \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}} \sum_{n'} \frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}} \hat{a}_{isn'}(\omega)\right).$$

Therefore, taking the variance,

$$\begin{aligned}
\mathrm{Var}(\hat{\ell}_{isn}(\omega)) &= \frac{1}{\left(\eta + \frac{1}{\kappa}\right)^2} \mathbb{E}\left[\left(\hat{a}_{isn}(\omega) - \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}} \sum_{n'} \frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}} \hat{a}_{isn'}(\omega)\right)^2\right] \\
&= \frac{1}{\left(\eta + \frac{1}{\kappa}\right)^2} \mathbb{E}\left[\hat{a}_{isn}(\omega)^2\right] \\
&\quad - 2\frac{1}{\left(\eta + \frac{1}{\kappa}\right)^2} \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}} \mathbb{E}\left[\hat{a}_{isn}(\omega) \sum_{n'} \frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}} \hat{a}_{isn'}(\omega)\right] \\
&\quad + \frac{1}{\left(\eta + \frac{1}{\kappa}\right)^2} \left(\frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}}\right)^2 \mathbb{E}\left[\left(\sum_{n'} \frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}} \hat{a}_{isn'}(\omega)\right)^2\right] \\
&= \frac{1}{\left(\eta + \frac{1}{\kappa}\right)^2} \left(\sigma_N^2 + \sigma_S^2\right) \\
&\quad - 2\frac{1}{\left(\eta + \frac{1}{\kappa}\right)^2} \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}} \left(\sigma_S^2 + \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \sigma_N^2\right) \\
&\quad + \frac{1}{\left(\eta + \frac{1}{\kappa}\right)^2} \left(\frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}}\right)^2 \left(\sigma_S^2 + \sum_{n' \in \mathcal{N}_{is}} \left(\frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}}\right)^2 \sigma_N^2\right).
\end{aligned}$$

We then aggregate up to the sectoral level,

$$\begin{aligned}
\sum_{n \in \mathcal{N}_{is}} \frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}} \mathrm{Var}(\hat{\ell}_{isn}(\omega)) &= \frac{1}{\left(\eta + \frac{1}{\kappa}\right)^2} \left(\sigma_N^2 + \sigma_S^2\right) \\
&\quad - 2\frac{1}{\left(\eta + \frac{1}{\kappa}\right)^2} \frac{\frac{1}{\nu} - \frac{1}{\kappa}}{\eta + \frac{1}{\nu}} \left(\sigma_S^2 + \sum_{n} \left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}\right)^2 \sigma_N^2\right)
\end{aligned}$$

$$+\frac{1}{\left(\eta+\frac{1}{\kappa}\right)^2}\left(\frac{\frac{1}{\nu}-\frac{1}{\kappa}}{\eta+\frac{1}{\nu}}\right)^2\left(\sigma_S^2+\sum_{n'\in\mathcal{N}_{is}}\left(\frac{\overline{\ell}_{isn'}}{\overline{\ell}_{is}}\right)^2\sigma_N^2\right)$$

$$=\left(\frac{1}{\eta+\frac{1}{\nu}}\right)^2\sigma_S^2+\left(\frac{1}{\eta+\frac{1}{\kappa}}\right)^2\sigma_N^2$$
$$-\left(\frac{1}{\nu}-\frac{1}{\kappa}\right)\frac{\eta+\frac{1}{\kappa}+\eta+\frac{1}{\nu}}{\left(\eta+\frac{1}{\kappa}\right)^2\left(\eta+\frac{1}{\nu}\right)^2}HHI_{is}\sigma_N^2.$$

The previous proposition then proves that the HHI is decreasing in N. □

C Robustness of Theory

In this section, we consider some extensions to the baseline model to show that the key theoretical takeaways are robust to alternate modeling assumptions.

C.1 Alternate Entry Condition

The free entry condition in the main text,

$$\psi_i=\frac{\mathbb{E}\left[\int_0^1\sum_{n\in\mathcal{N}_{is}}\pi_{isn}(\omega)ds\right]}{m_i},\tag{37}$$

imagines a staged entry game. All firms choose whether or not to enter the market, and then all of those firms are randomly assigned sectors and ex-ante productivity shocks. In this appendix, we consider an alternate entry game where a potential entrant sees the number of firms in each sector and the ex-ante shocks they have received and decides whether or not to enter. We will characterize this alternate free entry condition and prove that it is the same condition as the main text free entry condition (37).

Under the alternate entry game, the potential entrant can see that there is a mass $p(N,m_i)$ of sectors with N firms, where $p(N,m_i)=m_i^Ne^{-m_i}/N!$ is the probability mass function (pmf). Therefore, the probability that the firm enters a sector s with N firms already in it is $p(N,m_i)$. The firm internalizes that if it were to enter such a sector s, that sector would not have N firms, but instead $N+1$ firms. Nothing else about the equilibrium would change as that sector is small relative to the entire location so the expected profits of the firm would be the average profits of a firm in a sector with $N+1$ firms, $\pi^e_{i,N+1}\equiv\mathbb{E}[\pi_{isn}(\omega)|N_{is}=N+1]$, where expectations are taken over ex-ante and ex-post shocks.

Therefore, the free entry condition can be written,

$$\psi_i=\sum_{N=0}^{\infty}\pi^e_{i,N+1}p(N,m_i).\tag{38}$$

Now we will show that this free entry condition (38) is equivalent to (37). Taking the original free entry condition, notice that,

$$\frac{\mathbb{E}\left[\int_0^1 \sum_{n\in\mathcal{N}_{is}} \pi_{isn}(\omega)ds\right]}{m_i} = \frac{\sum_{N=0}^{\infty} \mathbb{E}\left[\sum_{n\in\mathcal{N}_{is}} \pi_{isn}(\omega)|N_{is}=N\right] p\left(N,m_i\right)}{m_i}$$
$$= \sum_{N=0}^{\infty} \pi^e_{i,N} \frac{N}{m_i} p(N,m_i).$$

Taking the alternate free entry condition, we have

$$\sum_{N=0}^{\infty} \pi^e_{i,N+1} p(N,m_i) = \sum_{N=0}^{\infty} \pi^e_{i,N+1} \frac{N+1}{m_i} \frac{m_i^{N+1} e^{-m_i}}{(N+1)!}$$
$$= \sum_{N=1}^{\infty} \pi^e_{i,N} \frac{N}{m_i} p(N,m_i) = \sum_{N=0}^{\infty} \pi^e_{i,N} \frac{N}{m_i} p(N,m_i),$$

proving the equivalence.

C.2 Labor Fixed Cost

In this section, we consider a version of the model where the fixed cost is paid in terms of local labor rather than the freely traded final good. The only substantive change is that the number of firms increases proportionately with the number of workers.

C.2.1 Environment

For the goods market to clear, total consumption must be equal to production,

$$c_i \ell_i = Y_i. \tag{39}$$

Labor can now be used in production or in paying the fixed cost to enter. In period 1, the worker decides how much labor to allocate to fixed cost investment, L_{if}, and sector s production, L_{is}. That is,

$$\boldsymbol{L_i} \in \mathcal{L} \equiv \left\{ \boldsymbol{L_i'} | L_{if} + \int_0^1 L'_{is} ds \leq 1 \right\}.$$

ψ_i labor is required for a firm to enter. Therefore, the demand for fixed cost labor is $\psi_i m_i$, and the fixed cost labor market clearing condition is that $L_{if}\ell_i = \psi_i m_i$.

For workers to allocate a non-zero amount of labor to fixed cost production, it must pay the average wage of the production sector, w_i. Therefore, the free entry condition is

$$\psi_i w_i = \frac{\mathbb{E}\left[\int_0^1 \sum_{n\in\mathcal{N}_{is}} \pi_{isn}(\omega)ds\right]}{m_i}.$$

C.2.2 Results

The regional production is unchanged except it is now a function of the mass of firms m_i and the amount of production labor $\ell_{ip} \equiv (\int_0^1 L_{is} ds)\ell_i$. Production workers are paid their marginal product so that,

$$w_i = \frac{(1-\eta)Y_i}{\ell_{ip}}.$$

Total profits are $Y_i - w_i \ell_{ip}$, so the free entry condition can be written,

$$\begin{aligned}
\psi_i w_i &= \frac{Y_i - w_i \ell_{ip}}{m_i} \\
&= \frac{\frac{1}{1-\eta} w_i \ell_{ip} - w_i \ell_{ip}}{m_i} \\
\psi_i &= \frac{\eta}{1-\eta} L_{ip} \frac{\ell_i}{m_i}.
\end{aligned}$$

This immediately implies that L_{ip} is constant and the mass of firms increases proportionally with population, as $m_i = (1 - L_{ip})\ell_i / \psi_i$. Nothing else about the model changes.

C.3 Firm Sorting

In this section, we extend the baseline to allow firm sorting. In particular, we allow firms to see their ex-ante productivity draw after paying a fixed cost of entry before deciding whether they want to pay the fixed cost to operate, as in Melitz (2003). This will introduce a new potential market failure, as large, productive firms know that their entry will have a disproportionate impact on the distribution of wages. This will distort the mass of firm entrants m_i and the cutoff value $\tilde{z}_i$ for which firms decide to enter.

C.3.1 Environment

Firms now make a two part entry decision. First, a firm may pay a fixed cost of ψ_i in terms of local labor to get an ex-ante productivit draw z_i from a pareto distribution $\mathcal{P}(\lambda, \zeta_i)$. We denote the mass of firms that pay this fixed cost by $\tilde{m}_i$. The firm can then decide whether to pay an operating cost ϕ in terms of local labor to enter. If the firm pays the operating cost, it is then randomly sorted into a sector. We will denote the mass of firms that decide to operate by m_i. Entry is distributed Poisson, just as in the main text. Firms then maximize profits taking prices and wages as given.

We will characterize the equilibrium in the quantitative version of the model in Section 3 so that we can assess how it changes our quantification in Appendix E.

In period 1, workers decide how much labor to allocate to investment, L_{if}, and sector s production, L_{is}. That is,

$$\boldsymbol{L_i} \in \mathcal{L} \equiv \left\{ \boldsymbol{L_i'} | L_{if} + \int_0^1 L_{is}' ds \leq 1 \right\}.$$

Demand for investment labor comes from the fixed cost of entering and the operating cost. The investment labor market clearing condition is that $L_{if}\ell_i = \psi_i \tilde{m}_i + \phi m_i$.

C.3.2 Characterizing the Entry Decision

We are unable to get qualitative results in this extended model, so instead, we characterize the equilibrium and simulate what that implies for the size of agglomeration effects. As we did in the main text, we will characterize firm decisions using backward induction.

In period 2, firms solve the problem,

$$\max_{\ell'} z_{isn} a_{isn}(\omega)(\ell')^{1-\eta} - w_{isn}(\omega)\ell'.$$

Solving this implies profits, in state of the world ω, are $\pi_{isn}(\omega) = \eta z_{isn}^{\frac{1}{\eta}} a_{isn}(\omega)^{\frac{1}{\eta}} w_{isn}(\omega)^{-\frac{1-\eta}{\eta}} (1-\eta)^{\frac{1-\eta}{\eta}}$. Taking a second order approximation around the point with no ex-post shocks implies that

$$\pi_{isn}(\omega) \approx \overline{\pi}_{isn}\left(1 + \frac{1}{\eta}\hat{a}_{isn}(\omega) - \frac{1-\eta}{\eta}\hat{w}_{isn}(\omega) + \frac{1}{2}\left(\frac{1}{\eta}\hat{a}_{isn}(\omega) - \frac{1-\eta}{\eta}\hat{w}_{isn}(\omega)\right)^2\right).$$

Taking expectations over ω implies average profits are,

$$\pi_{isn} = \eta z_{isn}^{\frac{1}{\eta}} \mathbb{E}\left[(z_{isn})^{\frac{1}{\eta}}\right]^{-(1-\eta)} \left(\frac{\ell_i^p}{m_i}\right)^{1-\eta} \tilde{g}\left(\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}, HHI_{is}\right).$$

where

$$\begin{aligned}\tilde{g}(x, H) \equiv 1 &+ \left(\frac{1}{2}\left(\frac{1+\frac{1}{\kappa}}{\eta+\frac{1}{\kappa}}\right)^2 - (1-\eta)\frac{1+\frac{1}{\kappa}}{\eta+\frac{1}{\kappa}}\frac{\frac{1}{\nu}-\frac{1}{\kappa}}{\left(\eta+\frac{1}{\kappa}\right)\left(\eta+\frac{1}{\nu}\right)}x\right. \\ &\left. + \frac{(1-\eta)^2}{2}\left(\frac{\frac{1}{\nu}-\frac{1}{\kappa}}{\left(\eta+\frac{1}{\kappa}\right)\left(\eta+\frac{1}{\nu}\right)}\right)^2 H\right)\sigma_N^2 \\ &+ \frac{1}{2}\frac{\eta+\frac{1}{\kappa}}{\eta+\frac{1}{\nu}}\left(1+\frac{1}{\nu}\right)^2\sigma_S^2.\end{aligned}$$

A firm with ex-ante productivity shock z will operate if its expected profits are higher than ϕ. To find that, one simply needs to find the firms expected labor share $\frac{\overline{\ell}_{isn}}{\overline{\ell}_{is}}$ and HHI, HHI_{is} given its ex-ante productivity shock.

We assume that expected profits are increasing in ex-ante productivity.[20] This implies a cutoff rule where firms above a cutoff $\underline{z}$ enter and those below do not. Define $\widetilde{HHI}(N, z, \underline{z})$ as the expected HHI of a sector with N firms, one of which has productivity z, and the cutoff productivity is $\underline{z}$. Similarly, define $\tilde{x}(N, z, \tilde{z})$ as the expected market share of a firm with productivity z in a sector

[20] We verify this assumption holds computationally.

with N firms and cutoff productivity $\tilde{z}$. In particular,

$$\widetilde{HHI}(N,z,\underline{z}) \equiv \mathbb{E}\left[\sum_{n\in\mathcal{N}_{is}}\left(\frac{(z_{isn}/\underline{z}_i)^{\frac{1}{\eta}}}{\sum_{n'}(z_{isn'}/\underline{z}_i)^{\frac{1}{\eta}}}\right)^2\middle|\underline{z}_i=\underline{z},z_{is1}=z\right]$$
$$\tilde{x}(N,z,\underline{z}) \equiv \mathbb{E}\left[\frac{(z_{is1}/\underline{z}_i)^{\frac{1}{\eta}}}{\sum_n(z_{isn}/\underline{z}_i)^{\frac{1}{\eta}}}\middle|\underline{z}_i=\underline{z},z_{is1}=z\right].$$

With a slight abuse of notation, define the functions $\widetilde{HHI}(m,z,\underline{z})$ and $\tilde{x}(m,z,\underline{z})$ as the expected values taking as given the mass of firms, before knowing what sector the firm is in. Formally, $\widetilde{HHI}(m,z,\underline{z}) \equiv \sum_{N=0}\widetilde{HHI}(N,z,\underline{z})\frac{N}{m}p(N,m)$, and $\tilde{x}(m,z,\underline{z}) \equiv \sum_{N=0}\tilde{x}(N,z,\underline{z})\frac{N}{m}p(N,m)$.

Combining these functions gives expected profits as a function of ex-ante productivity, the mass of firm entrants, the firm operating cutoff, and the number of production workers,

$$\tilde{\pi}\left(m,\ell_p,z,\underline{z}\right) \equiv \eta\left(\frac{\eta\lambda}{\eta\lambda-1}\right)^{-(1-\eta)} z^{\frac{1}{\eta}}\left(\underline{z}\right)^{-\frac{1-\eta}{\eta}}\left(\frac{\ell_p}{m}\right)^{1-\eta} g\left(m,z,\underline{z}\right),$$

where

$$\begin{aligned} g\left(m,z,\underline{z}\right) \equiv 1 &+ \left(\frac{1}{2}\left(\frac{1+\frac{1}{\kappa}}{\eta+\frac{1}{\kappa}}\right)^2 - (1-\eta)\frac{1+\frac{1}{\kappa}}{\eta+\frac{1}{\kappa}}\frac{\frac{1}{\nu}-\frac{1}{\kappa}}{\left(\eta+\frac{1}{\kappa}\right)\left(\eta+\frac{1}{\nu}\right)}\tilde{x}\left(m,z,\underline{z}\right)\right. \\ &\left. + \frac{(1-\eta)^2}{2}\left(\frac{\frac{1}{\nu}-\frac{1}{\kappa}}{\left(\eta+\frac{1}{\kappa}\right)\left(\eta+\frac{1}{\nu}\right)}\right)^2 \widetilde{HHI}\left(m,z,\underline{z}\right)\right)\sigma_N^2 \\ &+ \frac{1}{2}\frac{\eta+\frac{1}{\kappa}}{\eta+\frac{1}{\nu}}\left(1+\frac{1}{\nu}\right)^2\sigma_S^2, \end{aligned}$$

and we substituted in for $\mathbb{E}[(z_{isn})^{1/\eta}]$ using the fact that the cutoff is $\underline{z}$.

The operating productivity cutoff $\underline{z}_i$ is characterized by profits equaling the operating costs,

$$w_i\phi = \eta\left(\frac{\eta\lambda}{\eta\lambda-1}\right)^{-(1-\eta)}\underline{z}_i\left(\frac{\ell_{ip}}{m_i}\right)^{1-\eta} g\left(m_i,\underline{z}_i,\underline{z}_i\right). \tag{40}$$

The mass of operating firms is then the share of firms who paid the cost of entering willing to pay the operating cost,

$$m_i = (1-F(\underline{z}_i))\tilde{m}_i = \left(\frac{\zeta_i}{\underline{z}_i}\right)^{\lambda}\tilde{m}_i. \tag{41}$$

Denote by $\overline{\pi}_i$ average profits before a firm knows its ex-ante shock. Just as in the benchmark model, production workers are paid their marginal product. Therefore,

$$w_i = \frac{(1-\eta)Y_i}{\ell_{ip}}. \tag{42}$$

Total profits conditional on paying the intial entry costs are total production, minus the money paid for operating costs and the money paid to production workers, $\overline{\pi}_i = \frac{Y_i - w_i \ell_{ip} - w_i m_i \phi}{\tilde{m}_i}$. With free entry, $\overline{\pi}_i = w_i \psi_i$. Rearranging, one finds that,

$$(1-\eta)\ell_i = \ell_{ip} \tag{43}$$

$$\eta \ell_i = m_i \phi + \tilde{m}_i \psi_i. \tag{44}$$

To complete the description of equilibrium, we need to find production. Taking the same second order approximation to aggregate production as in the baseline implies that,

$$Y_i = \left((\tilde{z}_i)^{\frac{1}{\eta}} \frac{\eta\lambda}{\eta\lambda - 1} \right)^{\eta} \ell_{ip}^{1-\eta} m_i^{\eta} \left[1 + \frac{1}{2}(\sigma_S^2 + \sigma_N^2) + \frac{1-\eta}{2} \frac{1}{\eta + \frac{1}{\nu}} \sigma_S^2 + \frac{1-\eta}{2} \frac{\eta + \frac{1}{\nu} - \left(\frac{1}{\nu} - \frac{1}{\kappa}\right) HHI_i(m_i)}{\left(\eta + \frac{1}{\kappa}\right)\left(\eta + \frac{1}{\nu}\right)} \sigma_N^2 \right]. \tag{45}$$

Equations (40)-(45) jointly characterize equilibrium. In practice this implies that as population increases, m_i increases less than proportionately. Instead, $\underline{z}_i$ increases at the same time, increasing the average firm size as the location expands. We simulate the quantitative implications in Appendix E.

C.4 Including Capital

In this section, we extend the baseline model to include capital. There are a few ways to include capital. One could assume that capital is completely stuck in each firm across all states of the world. This is captured in our current model, where the entry costs include capital. Alternatively, we could introduce capital that is subject to movement costs just like labor. If the movement costs are the same as labor, then the model does not change at all.

In this subsection, we introduce freely traded capital and show that it is equivalent to our baseline model with an adjusted η. We show the quantitative implications in Appendix E.

C.4.1 Environment

We assume that Japan is a small open economy and can rent capital from international capital markets for an exogenous price r on spot markets in every state of the world. Firms have a Cobb-Douglas technology over labor and capital. They maximize profits by taking wages, the rental rate of capital, and prices as given,

$$\ell_{isn}(\omega), k_{isn}(\omega) \in \underset{\ell', k'}{\text{argmax}} \quad z_{isn} a_{isn}(\omega) \left((\ell')^{1-\alpha} (k')^{\alpha} \right)^{1-\eta} - w_{isn}(\omega)\ell' - rk'.$$

Denote by $k_i \equiv \int_0^1 \sum_{n \in \mathcal{N}_{is}} k_{isn}(\omega) ds$ the total capital used in location i. National market-clearing (24) is then adjusted to account for the goods that must be exported to pay for the capital used in

production,

$$\sum_{i\in\mathcal{I}} c_i\ell_i + rk_i + \psi_i m_i = \sum_{i\in\mathcal{I}} Y_i. \tag{46}$$

C.4.2 Equivalence

We show the equivalence between this economy and an economy with no capital in two steps. We start by solving for optimal capital taking as given the labor choice. We then plug this into the production function to get an adjusted production function only as a function of labor.

Capital solves

$$\max_{k'} z_{isn}a_{isn}(\omega)\ell_{isn}(\omega)^{(1-\alpha)(1-\eta)}(k')^{\alpha(1-\eta)} - rk'.$$

Taking the first-order condition, we find that,

$$rk_{isn}(\omega) = \alpha(1-\eta)y_{isn}(\omega).$$

Plugging this in, production is

$$\begin{aligned} y_{isn}(\omega) &= z_{isn}a_{isn}(\omega)\ell_{isn}(\omega)^{(1-\alpha)(1-\eta)}\left(\frac{\alpha(1-\eta)y_{isn}(\omega)}{r}\right)^{\alpha(1-\eta)} \\ &= (z_{isn}a_{isn}(\omega))^{\frac{1}{1-\alpha(1-\eta)}}\left(\frac{\alpha(1-\eta)}{r}\right)^{\frac{\alpha(1-\eta)}{1-\alpha(1-\eta)}}\ell_{isn}(\omega)^{\frac{(1-\alpha)(1-\eta)}{1-\alpha(1-\eta)}}. \end{aligned}$$

Furthermore, payments to labor are then

$$w_{isn}(\omega)\ell_{isn}(\omega) = (1-\alpha)(1-\eta)y_{isn}(\omega).$$

Then the economy is equivalent to one with no capital where the degree of decreasing returns to scale is $\tilde{\eta}$, defined by $1-\tilde{\eta} \equiv \frac{(1-\alpha)(1-\eta)}{1-\alpha(1-\eta)}$, and production is $\tilde{y}_{isn}(\omega) = (1-\alpha(1-\eta))y_{isn}(\omega)$. Payments to labor are then

$$w_{isn}(\omega)\ell_{isn}(\omega) = (1-\tilde{\eta})\tilde{y}_{isn}(\omega).$$

Furthermore, $rk_i = \alpha(1-\eta)Y_i$. Therefore, as $\tilde{Y}_i \equiv (1-\alpha(1-\eta))Y_i$, the national market-clearing condition can be rewritten as in our baseline model,

$$\sum_{i\in\mathcal{I}} c_i\ell_i + \psi_i m_i = \sum_{i\in\mathcal{I}} \tilde{Y}_i.$$

D Robustness of Estimation

D.1 Short-run across-firm labor supply by market size.

Here, we examine whether the estimated short-run labor supply elasticity across firms, κ, varies systematically with the size of the local labor market (LLM). In our main analysis, we showed that firms' employment responses to idiosyncratic shocks depend on their relative size within the market—larger firms expand more when productive because they can attract workers from other firms in the same market. This mechanism implies that what matters is a firm's size relative to its local labor pool, not the absolute size of the labor market itself.

Nonetheless, larger LLMs may exhibit stronger aggregate employment responses to wage changes for reasons unrelated to our mechanism, such as greater worker mobility, more diverse industry composition, or different adjustment frictions. To ensure that our results are not driven by such differences across locations, we re-estimate κ separately for large and small LLMs.

Table A1 reports the results. We classify LLMs based on their annual number of firms between 2002 and 2019, defining large LLMs as those that maintained at least 11 firms (i.e., more than 10) in every year of the sample period, and small LLMs as the remainder. This classification yields 1,339 large and 13,228 small markets out of 14,567 in total, and group assignments are fixed over time according to this criterion. Column (1) of Table A1 replicates the full-sample estimate from the main text, while Columns (2)–(3) show the results for the two subsamples.

The estimated elasticities are 1.38 for large markets and 2.33 for small markets, although the latter is not statistically significant due to the weak first stage. The difference between the estimates for large and small markets is not statistically significant, and the point estimate for large markets, which excludes the small markets, is smaller than the full-sample estimate. These results would work against our results, which suggest firms in larger markets expand more for the same shocks. Changing the cutoff around this threshold does not materially affect the results.

Table A1: Estimation of Short-run Labor Supply Elasticity across Firms: Subsample by Market Size

Dep. Var.: Log Employment Growth			
	All LLM	Large LLM	Small LLM
Log Wage Growth	1.62	1.38	2.33
	(0.70)	(0.68)	(1.76)
Observations	1,340,985	821,383	519,602
Unique Num. of LLMs	13,155	1,328	11,827
1st Stage F-Stat.	45.66	37.94	9.18

Note: This table shows the estimates of short-run labor supply elasticities across firms, κ, by subsamples. The specification is the same as Column (1) of the Table in the main text. Column (1) reports results for the full sample, which replicates the result in the main text. Columns (2)–(3) split local labor markets by annual firm counts over 2002–2019. Column (2) includes markets that maintained *at least 11 firms* (i.e., more than 10) in *every* year from 2002 to 2019 (i.e., $\min_{t\in[2002,2019]} \#\text{firms}_{ist} \geq 11$); Column (3) contains the remainder. Group assignments are fixed over time for each market based on this criterion. Exposure-robust shift-share standard errors are in parentheses. Market-year fixed effects are included in all the columns. Covariates include the sum of the share ('incomplete share') and its interaction with time fixed effects.

D.2 Shift-Share Instrument Diagnostic

In the main text, we drop the top 10 products with the largest absolute Rotemberg weights and find that the estimates are very similar, suggesting that the results are robust. Table A2 lists these 10 products, with Rotemberg weights, log growth in shipments, and the share of shipments. All values are averaged over the 2002–2018 base years.

Table A2: Top 10 Rotemberg Weight Products

Product Name	Rotemberg wt	Shift	Shipment (%)
Steelworks	0.267	0.023	0.31%
Iron and steel prepared products	-0.108	0.046	0.12%
Medical material preparations	0.073	-0.010	0.21%
Sushi and box lunches	-0.072	0.029	0.3%
Steel special-use shipbuilding	0.068	-0.002	0.03%
Semi-finished green tea	0.066	-0.000	0.02%
Part of semiconductor mfg equipment	0.063	0.054	0.11%
Fabrication of plastic film, leathers etc	-0.061	0.025	0.26%
Incandescent lamp fixtures	-0.060	0.047	0.07%
Misc. aliphatic intermediates	0.058	0.028	0.21%

Note: This table reports the top 10 (in absolute values) product categories according to the Rotemberg weights used when constructing shift-share instruments in (16). All values are averaged over the 2002–2018 base years. Shift is the log growth of shipments at the product level. Shipment (%) is the product shipment share in all products.

D.3 Labor Supply Elasticity over a Two-year Horizon

In the main text, we estimate labor supply elasticities (κ, ν) using one-year changes in employment and labor demand. Here, we estimate them using two-year changes. Tables A3 and A4 show the results. The estimates are larger over the longer horizon, as expected, since workers have more time to adjust. In particular, the baseline estimate of κ increases from 1.62 to 2.17, and the baseline estimate of ν increases from 0.80 to 1.33.

Table A3: Estimation of Short-run, Across-Firm Labor Supply Elasticity (2-year)

Dep. Var.: Log Employment Growth				
	(1)	(2)	(3)	(4)
Log Wage Growth	2.17	2.52	2.93	2.62
	(0.61)	(0.94)	(1.01)	(1.00)
LLM-Year FE	✓	✓	✓	✓
Control Lagged Emp Growth		✓	✓	✓
Drop Top 10 Rotemberg			✓	
Use Lagged Share				✓
1st-stage F Stat.	90.76	53.43	33.37	43.24
Num of Unique LLMs	9,337	9,111	9,080	9,111
Num of Unique Products (Shifts)	1,829	1,825	1,815	1,825
Num of Unique Firms	113,796	110,752	108,407	110,752
Num of Years	16	14	14	14
Shift-Year Obs.	28,866	25,144	25,004	25,140
Firm-Year Obs.	1,248,780	1,024,037	991,436	1,024,037

Note: This table shows the estimates of short-run labor supply elasticity across firms within markets, κ, following (15), using two-year changes. All columns include market-time fixed effects, the sum of the shares, and the interaction of the sum and time fixed effects as covariates. Columns (2), (3), and (4) include firm-level lagged log employment growth as an additional covariate. Column (3) drops the top 10 products (units of the shifts) in terms of the absolute values of Rotemberg weights, computed for the two-year shifts. Column (4) uses lagged shipment share instead of the time-invariant average shipment share to construct the shift-share IV in (16). Exposure-robust standard errors (Borusyak et al., 2025) are in parentheses.

Table A4: Estimation of Short-run, Across-Market Labor Supply Elasticity (2-year)

Dep. Var.: Log LLM Employment Growth				
	(1)	(2)	(3)	(4)
Log LLM Wage Growth	1.33	1.31	1.42	1.31
	(0.21)	(0.20)	(0.24)	(0.20)
Sector-Year FE	✓	✓	✓	✓
CZ-Year FE	✓	✓	✓	✓
Control Lagged Emp Growth		✓	✓	✓
Use Lagged Share			✓	
Weighted				✓
1st-stage F Stat.	215.25	229.35	183.97	994.14
Num of Unique LLM	12,218	11,612	11,612	11,612
Num of Years	16	14	14	14
LLM-Year Obs.	162,312	135,936	135,936	135,936

Note: This table shows the estimates of short-run labor supply elasticity across markets, ν, following (18), using two-year changes. The unit of observation is the local labor market (LLM)-year level. LLMs are defined as pairs of 3-digit sectors and commuting zones. All columns include sector-time fixed effects, cz-time fixed effects, the sum of the shares ('incomplete share'), and the interaction of the sum and time fixed effects as covariates. Columns (2), (3), and (4) add lagged LLM employment growth as a covariate. Column (3) uses lagged shipment share when constructing the firm-level shift-share IV in (16). Column (4) weights each local labor market using the median total employment over time. Exposure-robust standard errors (Borusyak et al., 2025) are in parentheses.

D.4 Summary Statistics for the Baseline Reduced-form Regression Sample

Table A5 reports summary statistics for the exact estimation sample used in Column (1) of Table 4. We report the distributions separately for the beginning and end of the sample period using 2002 and 2018 as the base years.

Table A5: Summary Statistics: Baseline Reduced-form Regression Sample

	2002						2018					
	Mean	p10	p25	p50	p75	p90	Mean	p10	p25	p50	p75	p90
Establishment-level data												
Employment	62.5	11.0	15.0	24.0	50.0	117.0	71.9	11.0	15.0	26.0	57.0	137.0
Employment Growth (in log)	-0.007	-0.154	-0.069	0.000	0.054	0.143	-0.020	-0.154	-0.069	0.000	0.038	0.102
SSIV Shock (in log)	0.010	-0.070	-0.021	0.005	0.039	0.087	-0.012	-0.069	-0.031	-0.003	0.013	0.032
Observations	81,784						78,900					
LLM-level data												
Total Employment	481.2	18.0	42.0	131.0	404.0	1,102.0	538.2	19.0	44.0	139.5	430.0	1,176.0
Num. of Establishments	7.7	1.0	1.0	3.0	7.0	17.0	7.5	1.0	1.0	3.0	7.0	17.0
Payroll HHI	5,679	1,323	2,606	5,099	10,000	10,000	5,734	1,373	2,667	5,155	10,000	10,000
Observations	10,618						10,540					

Note: This table reports summary statistics for the exact estimation sample used in Column (1) of Table 4. Establishment-level changes use base-year-to-next-year changes, so 2002 corresponds to 2002–2003 and 2018 corresponds to 2018–2019. Local labor market (LLM) statistics aggregate sampled establishments within each LLM-year, where LLMs are defined as 3-digit industry by commuting zone cells.

D.5 Reduced-form Results with Various Fixed Effects

In Table 4 in the main text, we show the baseline reduced-form results. Here, we show how those results change when we vary the fixed effects in Table A6. The negative interaction term remains when we include year fixed effects or CZ-year fixed effects, but it becomes small and statistically insignificant once we include sector-year or local labor market-year fixed effects. This is exactly what we would expect, as those fixed effects absorb the variation in shocks across sectors and local labor markets that identify the mechanism.

Table A6: Responses of Employment to Product-level Shocks (Various Fixed Effects)

	Dep. Var.: Log Employment Growth			
	(1)	(2)	(3)	(4)
Shock	0.048	0.046	0.025	0.021
	(0.005)	(0.005)	(0.005)	(0.005)
Payroll Share	-0.023	-0.023	-0.042	-0.396
	(0.005)	(0.005)	(0.005)	(0.014)
Shock x Payroll Share	-0.028	-0.028	-0.004	0.000
	(0.010)	(0.010)	(0.010)	(0.019)
Implied Ratio	-0.586	-0.598	-0.145	0.003
95% CI	[-0.929, -0.243]	[-0.956, -0.240]	[-0.893, 0.602]	[-1.739, 1.745]
Firm FE	✓	✓	✓	✓
Year FE	✓			
CZ-Year FE		✓		
Sector-Year FE			✓	
LLM-Year FE				✓
Num of Unique LLMs	12,968	12,968	12,967	9,098
Num of Unique Firms	111,508	111,508	111,508	109,122
Num of Years	17	17	17	17
Firm-Year Obs	1,391,787	1,391,787	1,391,775	1,336,245

Note: This table shows the estimates of the impact of a firm-level demand shock on employment, following (23). All columns include firm fixed effects. All columns also include the sum of the shares ('incomplete shares') and its interaction with the payroll share, both interacted with time fixed effects, as covariates. Column (1) includes year fixed effects. Column (2) includes CZ-year fixed effects. Column (3) includes sector-year fixed effects. Column (4) includes local labor market (LLM)-year fixed effects. The implied ratio is the ratio of the coefficient on the interaction term between the shock and payroll share within local labor markets, β_3, to the coefficient on the shock, β_1. The 95% confidence interval of that implied ratio is also reported. Exposure-robust standard errors (Borusyak et al., 2025) are in parentheses for the shock and interaction terms. Standard errors for payroll share are clustered at the firm level.

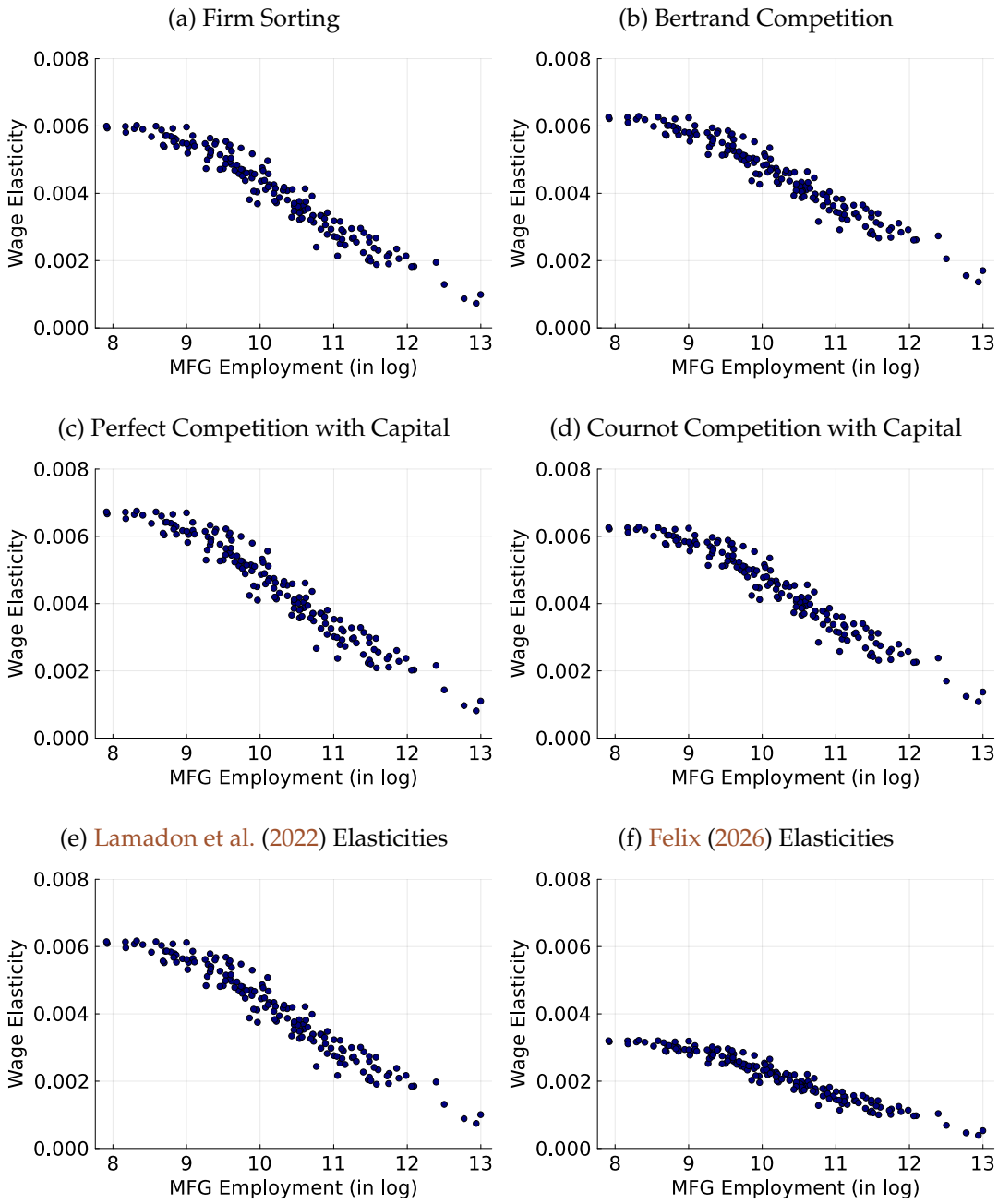
E Robustness of Quantification

Figure A7 shows wage elasticities across alternative specifications. Panel (a) shows the results with firm sorting. The elasticity at the smallest commuting zone becomes 0.006, which is slightly lower than the baseline (0.007). Panel (b) presents the results under Bertrand Competition. The qualitative conclusions are unchanged. The wage elasticities are slightly smaller than under either of the other two conduct assumptions. In Panels (c) and (d), we present the results with freely traded capital.[21] Panels (e) and (f) use the labor market elasticity estimates in the US (Lamadon et al., 2022) and Brazil (Felix, 2026). The wage elasticities are all slightly smaller than the baseline, except for Panel (f), where we use the labor market elasticity estimates in Brazil.[22] The patterns across commuting zones and the differences across conduct assumptions remain unchanged.

[21] We use the alternate calibration with $1 - \tilde{\eta} = \frac{(1-\alpha)(1-\eta)}{1-\alpha(1-\eta)}$, where we take α to match the capital share in Japan.

[22] The magnitude of our estimates depends on the differences between κ and ν, and Felix (2026) reports that they are not statistically different in Brazil.

Figure A7: Wage Elasticities (Alternative Settings)



Note: Panels show the wage elasticity against the natural logarithm of total employment at manufacturing establishments in the commuting zone in 2019. Each commuting-zone observation represents one of the Japanese commuting zones in the sample. Panel (a) includes firm sorting. Panel (b) assumes Bertrand competition in the labor market. Panels (c) and (d) include capital. Panel (c) assumes perfect competition, and panel (d) assumes Cournot competition. Panel (e) uses the estimates of the labor market elasticities κ and ν from Lamadon et al. (2022) for the US, while Panel (f) uses those from Felix (2026) for Brazil.